QUALIFICATION STANDARD AND PLAN

Advanced Acoustic Analysis Course (AAAC)

NQUAL – AEJR

MITE ID – 123261

RECORD OF AMENDMENTS

The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC(A)) is the designated development centre responsible for the maintenance of training. Amendments to Performance Objectives (POs) are to be requested through NTDC(A) and approved by CNPTG. Amendments to Lesson Guidance (LG) are to be approved by the Commanding Officer (CO) of NTDC(A).

Qualification Standard and Plan (QSP) versions with a 1 preceding the decimal indicate legacy 4 Chapter documents. QSP versions with a 2 preceding the decimal indicate Firefly 2 Chapter documents. QSP versions with a 3 preceding the decimal indicate 3 Chapter documents. Versions with a number followed by a bracketed number indicate a stray from this nomenclature. The number in brackets indicates the corrected version number.

Version Number	Amendment Type	Approval Date	Authority
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S & P – V2		18 June 2008	TRINITY(ADAC)
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V3 Ch 3&4		31 October 2011	TRINITY (ADAC)
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TABLE OF CONTENTS

RECORD OF AMENDMENTS ..... **I**

TABLE OF CONTENTS ..... **II**

ACRONYMS AND ABBREVIATIONS ..... **IV**

FOREWORD ..... **VII**

CHAPTER 1: TRAINING MANAGEMENT DETAILS **1**

AIM 2
COURSE DURATION 2
REQUIRED COURSE STAFFING 2
COURSE CAPACITY 3
PREREQUISITES 3
PRIOR LEARNING ASSESSMENT AND RECOGNITION 3
RELATED DOCUMENTS 4
COURSE LOADING 4
SPECIAL INSTRUCTIONS 4

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE **5**

PO 001 – ANALYZE ACOUSTIC DATA 6
EO 001.01 – Interpret Passive Aural Listening (PAL) 8
LG 001.01.01 – Recognize Aural Acoustic Characteristics 8
LG 001.01.02 – Recognize Aural Transient Events 10
LG 001.01.03 – Analyze Aural Acoustic Data 11
LG 001.01.04 – Classify Contacts Aurally 13
EC 001.01 – Interpret Passive Aural Listening (PAL) 14
EO 001.02 – Conducting Acoustic Gram Analysis 16
LG 001.02.01 – Analysis approach 16
LG 001.02.02 – Identify Acoustic Sources 18
LG 001.02.03 – Analyzing Diesel Submarines 21
LG 001.02.04 – Analyzing Non-Allied Nuclear Submarines 22
LG 001.02.05 – Analyzing Allied Nuclear Submarines 23
LG 001.02.06 – Analyzing Naval Surface Vessels 24
LG 001.02.07 – Classify Passive Acoustic Contacts 25
EC 001.02 – Conducting Acoustic Gram Analysis 28
EO 001.03 – Conduct Tactical Analysis 30
LG 001.03.01 – Interpret Tactical Events 30
LG 001.03.02 –Tactical Analysis of Diesel submarines 31
LG 001.03.03 –Tactical Analysis of Non-Allied Nuclear submarines 32
LG 001.03.04 –Tactical Analysis of Allied Nuclear submarines 33
LG 001.03.05 –Tactical Analysis of Naval Surface Vessels 34
LG 001.03.06 – Rapid Tactical Analysis 35
EC 001.03 – Conduct Tactical Analysis 36
PC 001 – Analyze Acoustic Data 38

CHAPTER 3: ASSESSMENT OF TRAINEE **40**

GENERAL 41
ENABLING AND PERFORMANCE CHECKS 41
RESPONSIBILITY FOR TESTING 41
PROGRESS MONITORING 41
TRAINING REVIEW BOARD 42

COURSE REPORTS 42
ANNEX A – REFERENCES 1
ANNEX B – TASK LIST 1
ANNEX C – RECORD OF DECISIONS 1
ANNEX D – TASK ANALYSIS DIAGRAM 1
ANNEX E – COSTING INFORMATION 1
ANNEX F – ASSESSMENT PLAN 1
ANNEX G – TEL DEVELOPMENT OPPORTUNITIES 1
ANNEX H – SCHEDULING INFORMATION 1
RECOMMENDED SCHEDULE 3
ANNEX J – EVALUATION MATRIX 1

ACRONYMS AND ABBREVIATIONS

A	
AC	Alternating Current
ADAC	Acoustic Data Analysis Centre
ADC	Acoustic Deception Countermeasure
AES	Airborne Electronic Sensors
AIP	Air Independent Propulsion
ARS	Acoustic Response Sheet
ASO	Acoustic Sensor Operator
ATT	Actual Training Time
B	
BAT	Base Administrative Time
BB	Broadband
C	
CAF	Canadian Armed Forces
CCA	Curriculum Control Authority
CEO	Canadian Eyes Only
CFCD	Canadian Forces Classified Document
CFITES	Canadian Forces Individual Training and Education System
Cmdt	Commandant
CNPTG	Commander Naval Personnel and Training Group
CO	Commanding Officer
CPA	Closest Point Of Approach
CSNI	Consolidated Secret Network Infrastructure
D	
DAOD	Defence Administrative Orders and Directives
DARCD	Determine, Analyse, Relate, Correlate, Decide
DC	Direct Current
DEMON	Demodulated Noise
DSCDS	Defence Subject Classification and Disposition System
DND	Department of National Defence
DPGR	Director Personnel Generation Requirements
DWAN	Defence Wide Area Network
E	
EC	Enabling Check
EDI	Explain, Demonstrate, Imitate
EER	Explosive Echo Ranging
EO	Enabling Objective
ESUS	Electronic Signal Underwater Sound
G	
GP Net	General Purpose Network
H	
HMS	Hull Mounted Sonar
I	
IAW	In Accordance With

ID	Identification
IMS	Information Management Standard
IOT	In Order To
IS	Information Systems
IT	Information Technology
L	
LFA	Low Frequency Active
LO	Learning Objective
LOFAR	Low Frequency Analyzer and Ranging
LS	Lesson Specification
LG	Lesson Guidance
M	
MITE	Military Individual Training and Education
MOAT	Maritime Operational Aircrew Training
MOSID	Military Occupational Structure Identification Code
N	
NAEB	Naval Anti-Escort Beacon
NAVORD	Naval Order
NQual	National Qualification
NFS	Naval Fleet School
NFS(A)	Naval Fleet School Atlantic
NFS(P)	Naval Fleet School Pacific
NOPF	Naval Oceanic Processing Facility
NPTG	Naval Personnel and Training Group
NPTORD	Naval Personnel and Training Order
NTDC(A)	Naval Training Development Centre (Atlantic)
NTI	Naval Training Instructions
O	
OSN	Own Ship Noise
OSS	Occupational Specialty Specification
P	
PAL	Passive Aural Listening
PC	Performance Check
PO	Performance Objective
PLAR	Prior Learning Assessment and Recognition
PRB	Performance Review Board
PT	Physical Training
Q	
QSP	Qualification Standard and Plan
QS	Qualification Standard
R	
R&Q	Rations & Quarters
RCAF	Royal Canadian Air Force
RCN	Royal Canadian Navy
RPM	Revolutions Per Minute

RTA	Rapid Tactical Analysis
S	
SAT	Satisfactory
SO	Standing Order
SONAR	Sound Navigation And Ranging
SOP	Standard Operating Procedure
SRC	Speed Related Component
SS	Specialty Specification
SSMA	Ships Service Motor Alternator
SSMG	Ships Service Motor Generator
SSTA	Ships Service Turbo Alternator
SSTG	Ships Service Turbo Generator
SUS	Signal Underwater Sound
T	
TACCO	Tactical Communications Officer
TAT	Training Administrative Time
TA	Training Authority
TD	Temporary Duty
TDO	Training Development Officer
TE	Training Establishment
TEL	Technology Enabled Learning
TIOW	Transmission Interval Of the Watch
TKT	Threshold Knowledge Test
TO	Training Officer
TP	Training Plan
TPK	Turns Per Knot
TRB	Training Review Board
TS	Training Site
TTT	Training Travel Time
U	
UNSAT	Unsatisfactory
V	
Vol	Volume

FOREWORD

1. This training is based on the specification found at Director Personnel Generation Requirements (DPGR) website: **http://cmp-cpm.mil.ca/en/support/military-personnel/dpgr-index.page**.

2. A trainee will be granted the National Qualification (NQual) AEJR when the standards listed in all POs have been met.

3. A trainee will be granted the Military Individual Training and Education (MITE) identification (ID) 123261 when standards listed in all POs have been met.

4. The designated TE for this training include:

   a. TRINITY ADAC; and

   b. Naval Fleet Schools (Atlantic and Pacific).

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this QSP is to establish the requirements and training for the Advanced Acoustic Analysis Course (AEJR). Possession of this specialty will enable personnel to perform advanced passive acoustic analysis, interpret Passive Aural Listening, conduct tactical analysis and report acoustic intelligence.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) Naval Personnel and Training Order (NPTORD) 4500-22 Annex B Table 3.1. Partial days are rounded up. For this course, a period of instruction is 50 minutes in duration and each day has 8 instructional periods. The course duration calculation is based on maximum load.

Course Duration Calculation		
Allocation of Actual Training Time (ATT)	Periods	Days
Lesson Guidance (LG)	125	15.625
Enabling Checks (ECs)	10	1.25
Performance Checks (PCs)	5	0.625
ATT SUB TOTAL:	**140**	**17.5 Days**
Allocation of BAT, TAT, TTT and PT	Periods	Days
Base Administrative Time (BAT)	2	0.25
Training Administrative Time (TAT)	2	0.25
Training Travel Time (TTT)	0	0
Physical Fitness (PT)	12	1.5
BAT, TAT, TTT, PT SUB TOTAL:	**16**	**2**
***Partial days are rounded up** **TOTAL:**		**20**

REQUIRED COURSE STAFFING

3. The following personnel, including guest lecturers, are required to conduct this training:

Position	Rank	Quantity	Qualification(s)	Security Requirement(s)	PO/EO/LG
Training Officer	Civ	1	AEJR, AHCH, AHCJ	Level III TOP SECRET	All POs
Senior Instructor	PO2/MS - Civ	2	AEJR, AHCH,	Level II SECRET	All POs
PAL Instructor	PO2-MS	1	AEJR, AHCH, AILO, AILS	Level II SECRET	EO 001.01

Critical Training Resources	No. of Days
Classroom with up to 12 Osprey workstations	20
Audio System with the following specifications: 20Hz-20kHz frequency response, Analog Graphic equalizer	20

COURSE CAPACITY

4. Planning for the conduct of training is based on availability of resources; course capacities are as follows:

a. Maximum – 12;

b. Minimum – 4; and

c. Optimum – 9.

PREREQUISITES

5. Personnel must meet the following requirements to be selected for this specialty::

a. For Officers, MPA - have completed MOAT followed by 6 months operational experience or Maritime Helicopter - be ADJJ qualified;

b. For NCMs, be RQ – S1 qualified and have 6 months of subsequent operational experience;

c. Be selected for employment as an ASO in a Maritime Patrol Aircraft, SENSO in a Maritime Helicopter, staff Acoustic analyst in ADAC/DIAC/NOPF or other passive acoustic related positions;

d. Possess a security clearance of Level II or higher.

6. Due to the level of instruction, Training Establishments (TEs) reserve the right to conduct a Threshold Knowledge Test (TKT) on potential candidates to determine whether they have the prerequisite skills for attendance. TKTs may also be administered upon request from units.

PRIOR LEARNING ASSESSMENT AND RECOGNITION

7. Prior Learning Assessment and Recognition (PLARs) shall be conducted IAW Naval Order (NAVORD) 4500-2, Defence Administrative Orders and Directives (DAOD) 5031-1, and Canadian Forces Individual Training and Education System (CFITES) Volume (Vol) 12.

RELATED DOCUMENTS

8. The following documents are related to this QSP:

   a. SQ - Advanced Acoustic Analyst;

   b. QSP – RQ-MS SONAR Op;

   c. QS - Acoustic Sensor Operator; and

   d. TP - Acoustic Sensor Operator

COURSE LOADING

9. Priority for student loading on individual course serials shall be as follows:

   a. RQ-MS SONAR Operator candidates (up to six per serial);

   b. AES Operator (LRP and MH) students on operational squadrons;

   c. Other SONAR Ops, and AES Ops based on job requirements or career development; and

   d. Allied personnel; preference will be given to agencies similar to TRINITY.

SPECIAL INSTRUCTIONS

10. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 – ANALYZE ACOUSTIC DATA

1. **Performance**: Analyze Acoustic Data (T0001).

2. **Conditions**:

   a. Given:

      (1) Acoustic processor;
      (2) Calculator;
      (3) References;
      (4) Chronoscope/stopwatch;
      (5) Passive acoustic data;
      (6) Acoustic response sheet; and
      (7) Headphones.

   b. Denied: Nil; and

   c. Environment: All conditions where Advanced Acoustic Analysts are employed.

3. **Standard**: The Advanced Acoustic Analyst shall Analyze Acoustic Data (T0001), IAW refs, by:

   a. Interpret Passive Aural Listening (T0002), by:

      (1) Recognizing aural acoustic characteristics;
      (2) Recognizing aural transient events;
      (3) Analyzing aural acoustic data; and
      (4) Classifying contacts aurally.

   b. Conducting acoustic gram analysis, to include:

      (1) Identifying acoustic sources (T0009);
      (2) Conducting contact correlation (T0008);
      (3) Interpreting acoustic sources (T0007);
      (4) Classifying passive acoustic contacts (T0010); and
      (5) Compiling acoustic intelligence data (T0011).

   c. Conduct tactical analysis (T0003), by:

      (1) Interpreting dynamic events (T0004);
      (2) Interpreting tactical implications of passive acoustic data (T0005); and
      (3) Interpreting changes in contact's readiness/ tactical states (T0006).

4. **Training Limitations**: Nil.

EO 001.01 – INTERPRET PASSIVE AURAL LISTENING (PAL)
LG 001.01.01 – RECOGNIZE AURAL ACOUSTIC CHARACTERISTICS

LG 001.01.01 – Recognize Aural Acoustic Characteristics		
Location Conditions	**Instructional Sequence**	
Classified Classroom	Beginning of course	
Time	**Classification**	**Instructor**
450 minutes	Secret	AAAC Instructor
Learning Objectives	**Instructional Method(s)**	
1) Acoustic Processor functionality 2) Aural propulsion sources 3) Prime movers 4) Aural Weapon sources	1) Explain, Demonstrate, Imitate (EDI) 2) Lecture 3) Practical	
Scenario	Nil	
Activity Steps	1) Students are provided with their own Osprey workstation. Instructor to conduct EDI on how to use Osprey system to include; a. Doppler tool b. Harmonic cursor c. Annotation tool d. Audio functions e. Aspect f. Invert function g. Ratio cursor h. Floating cursor i. Lloyd's Mirror (50 minutes – Method 1) 2) Instructor will give lecture on the following with audio examples; a. Shaft rate b. Blade count c. Blade rate d. Varying shaft RPM e. Multiple shafts Students will on their own identify each of the sources and will discuss as a class afterwards.(200 minutes – Method 2 and 3) 3) Instructor will give lecture on the following with audio examples of the following; a. Compressed cavitation b. Blade resonance c. Shaft rub d. Singing screw e. Blade flutter f. Blade slap g. Blade in/out	

LG 001.01.01 – Recognize Aural Acoustic Characteristics	
	Students will identify each of the sources on their own, and will discuss as a class afterwards. (50 minutes – Method 2 and 3) 4) Instructor will give lecture on the following with audio examples of the following; a. Diesel audible b. Exhaust flutter c. Whine d. Battery e. Main Motor Hum Students will on their own identify each of the sources and will discuss as a class afterwards. (50 minutes – Method 2 and 3) 5) Instructor will give lecture on the following with audio examples of the following; a. Torpedo b. Missiles c. Launch sequence Students will on their own identify each of the sources and will discuss as a class afterwards. (100 minutes – Method 2 and 3)
Instructor Resources	**References**
1) Sound system 2) IT support 3) PAL Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Chronoscope-Stop watch 7) Osprey workstation- 1 per student 8) PowerPoint lecture on OSPREY system 9) PowerPoint lecture with audio file examples of acoustic sound sources	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual 5) ADAC Acoustic Response Sheet Marking Guide
Note(s) to Instructor	
1) During activity 1, more attention to be focused on students with lack of experience. 2) Instructor may use any additional audio files as examples.	

EO 001.01 – INTERPRET PASSIVE AURAL LISTENING (PAL)
LG 001.01.02 – RECOGNIZE AURAL TRANSIENT EVENTS

LG 001.01.02 – Recognize Aural Transient Events		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After LG 001.01.01	
Time	**Classification**	**Instructor**
100 minutes	Secret	AAAC Instructor
Learning Objectives	**Method(s)**	
1) Aural Transient sources	1) Lecture 2) Practical	
Scenario	Nil	
Activity Steps	1) Instructor will give lecture on the following with audio examples of the following; a. Hydraulics b. Hydrodynamic c. Machinery d. Foghorn e. Baby Cry f. Babbling brook g. Frying fat h. Rasping i. Laughing Man / Big Bee j. Casing Rattle Students will on their own identify each of the sources and will discuss as a class afterwards. (100 minutes – Method 1 and 2)	
Instructor Resources	**References**	
1) Sound system 2) IT support 3) PAL Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Chronoscope-Stop watch 7) Osprey workstation- 1 per student 8) PowerPoint lecture with audio file examples of acoustic sound sources	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual	
Note(s) to Instructor		
1) Instructor may use any additional audio files as examples.		

EO 001.01 – INTERPRET PASSIVE AURAL LISTENING (PAL)
LG 001.01.03 – ANALYZE AURAL ACOUSTIC DATA

LG 001.01.03 – Analyze Aural Acoustic Data		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After LG 001.01.02	
Time	**Classification**	**Instructor**
250 minutes	Secret	AAAC Instructor
Learning Objectives	**Method(s)**	
1) Manmade aural sources 2) Aural biological sources 3) Aural active transmissions 4) Environmental noise	1) Lecture 2) Practical	
Scenario	Nil	
Activity Steps	1) Instructor will give lecture on the following with audio examples of the following; a. Trawl sequence b. Mooring Buoys c. Seismic profiling d. Oil Rigs e. Aircraft f. Sonobuoy deployment g. SUS and ESUS h. ADC i. NAEB j. Under Water Communications k. Gun Fire Students will on their own identify each of the sources and will discuss as a class afterwards. (50 minutes – Method 1 and 2) 2) Instructor will give lecture on the following with audio examples of the following; a. Marine mammals b. Shrimp c. Fish Students will on their own identify each of the sources and will discuss as a class afterwards. (50 minutes – Method 1 and 2) 3) Instructor will give lecture on the following with audio examples of the following; a. HMS b. Active Sonobuoy c. LFA d. EER e. Pulse Length f. Pulse Width g. TIOW	

LG 001.01.03 – Analyze Aural Acoustic Data	
	h. Classification Students will on their own identify each of the sources and will discuss as a class afterwards. (100 minutes – Method 1 and 2) 4) Instructor will give lecture on the following with audio examples of the following; a. Rain b. Ice c. Coastal noise d. Seismic e. Geologic noise f. Sea state Students will on their own identify each of the sources and will discuss as a class afterwards. (50 minutes – Method 1 and 2)
Instructor Resources	**References**
1) Sound system 2) IT support 3) PAL Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Chronoscope-Stop watch 7) Osprey workstation- 1 per student 8) PowerPoint lecture with audio file examples of acoustic sound sources	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor	
1) Instructor may use any additional audio files as examples.	

EO 001.01 – INTERPRET PASSIVE AURAL LISTENING (PAL)
LG 001.01.04 – CLASSIFY CONTACTS AURALLY

LG 001.01.04 – Classify Contacts Aurally		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After LG 001.01.03	
Time	**Classification**	**Instructor**
250 minutes	Secret	AAAC Instructor
Learning Objectives	**Method(s)**	
1) Analyze and Classify PAL Mixed Labs	1) Practical 2) Group Debrief	
Scenario	Nil	
Activity Steps	1) Instructor will provide students with 1 mixed lab. Students will conduct Passive Aural Listening analysis and provide answers on PAL ARS. (30 minutes – Method 1) 2) Class will conduct group debrief, taking up answers and discussing content. Each student to mark their own ARS. (20 minutes – Method 2) 3) Students will continue conducting a minimum additional 4 mixed labs with group debriefs. (200 minutes – Method1 and 2)	
Instructor Resources	**References**	
1) Sound system 2) IT support 3) PAL Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Chronoscope-Stop watch 7) Osprey workstation- 1 per student 8) Audio Wave Files 9) Mixed Labs (Minimum 5)	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual	
Note(s) to Instructor		
1) Students may be given additional Mixed Labs if time permits.		

EO 001.01 – INTERPRET PASSIVE AURAL LISTENING (PAL)
EC 001.01 – INTERPRET PASSIVE AURAL LISTENING (PAL)

EC 001.01 – Interpret Passive Aural Listening (PAL)		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.01	Classified Classroom	After LG 001.01.04
Time	**Classification**	**Instructor**
150 minutes	Secret	AAAC Instructor
Scenario	Nil	
Assessment Steps	**Input** – Student is given a mixed lab with contact audio files. **Performance** – Student will interpret Passive Aural Listening analysis on the contacts provided, fill out the PAL ARS and provide contact classification. (100 minutes) **Output** – Trainee must attain 75% of both analysis and classification to pass this assessment. **Feedback** –Student is provided written feedback via the ARS. Instructor conducts class debrief. Students may discuss answers with instructor. Instructor will have 1 hour to grade ARS. (50 minutes)	
Instructor Resources		
1) Sound system 2) IT support 3) PAL Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Chronoscope-Stop watch 7) Osprey workstation- 1 per student 8) Audio Wave Files 9) Mixed Lab		
Note(s) to Instructor		
1) EC 001.01 Practical assessment is designed to evaluate the trainee's ability to interpret PAL sources. 2) This test must consist of .Wav files representing a cross section of the material instructed. 3) Pass mark will be 75%. 4) References are permitted.		
Assessment Limitations	Nil	
Impact of Failure	Failure to achieve the standard on this Critical EC will result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this EC, continuation of training without a re-write may be recommended. See Chapter 3: Assessment of Trainee for further details.	

Passive Aural Listening Acoustic Response Sheet Example

# Shaft	Single	1 pt	# Blades	1pt	SRPM	1-2 pts
	Multiple	1 pt				
Characteristics & Transients					Classification	
1 per characteristic IAW Master					2 pts	

EO 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS
LG 001.02.01 – ANALYSIS APPROACH

LG 001.02.01 – Analysis Approach		
Location Conditions		**Instructional Sequence**
Classified Classroom		After EC 001.01
Time	**Classification**	**Instructor**
200 minutes	Secret	AAAC Instructor
Learning Objectives		**Method(s)**
1) Method of Acoustic Analysis 2) Timeline Analysis 3) Acoustic Response Sheet (ARS) 4) Intelligence Update		1) Lecture 2) EDI 3) Guest Lecture
Scenario	Nil	
Activity Steps	1) Instructor to conduct review of Analysis method and acoustic source characteristics to include; a. Broadband/Narrowband b. Discrete/Diffuse c. Stable/Unstable d. Timeline analysis (50 minutes – Method 1 and 2) 2) Instructor to conduct review of Timeline Analysis method to include; a. Steps to analysis i. D.A.R.C.D. ii. PAL iii. Contact correlation iv. Interference patterns v. Colour wheel vi. Dynamic events vii. CPA viii. TPK vs. Doppler speed ix. Signature characteristics b. Class indicators c. Class Identifiers (50 minutes – Method 1 and 2) 3) Instructor to conduct a lecture on the Acoustic Response Sheet (ARS) marking guide and the use of the Acoustic Intelligence Guide (AIG). (50 minutes – Method 1 and 2) 4) TRINITY Intelligence will provide a Guest Lecturer to conduct a presentation on Real World current submarine and World Order of Battle intelligence update. (50 minutes – Method 4)	

LG 001.02.01 – Analysis Approach	
Instructor Resources	**References**
1) Sound system 2) IT support 3) Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Chronoscope/Stop watch 7) Osprey workstation- 1 per student 8) Audio Wave Files 9) TRINITY Intelligence Guest Lecture presentation	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor	
1) Instructor will be responsible for liaising with TRINITY Intelligence to schedule the Guest Lecture. This lecture can be given anytime during the course for flexibility with the schedule.	

EO 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS
LG 001.02.02 – IDENTIFY ACOUSTIC SOURCES

LG 001.02.02– Identify Acoustic Sources		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After LG 001.02.01	
Time	**Classification**	**Instructor**
600 minutes	Secret	AAAC Instructor
Learning Objectives	**Instructional Method(s)**	
1) Propulsion related sources 2) Electrical sources/ Hydraulic pumps 3) Weapon Mechanics 4) Decoys and Jammers 5) Transients 6) Hydrodynamic sources 7) Active SONAR Sources 8) Merchant Vessel sources	1) Lecture 2) EDI 3) Practical Lab	
Scenario	Nil	
Activity Steps	1) Instructor to explain and demonstrate via PowerPoint lecture on Propulsion Sources to include; a. Propulsors b. Propellers c. Blade d. Shaft e. Harmonics f. DEMON (100 minutes – Method 1) 2) Instructor to explain and demonstrate via PowerPoint lecture on Continuously / Intermittently operated sources to include; a. SSTG/ SSTA/ b. SSMA/ SSMG i. Pole motors ii. Synchronous motors iii. Squirrel cage induction motor iv. Slip rates c. Harmonics d. AC Auxiliaries e. DC Auxiliaries f. Pumps g. Blowers h. Fans (50 minutes – Method 1)	

LG 001.02.02– Identify Acoustic Sources
3) Instructor to explain and demonstrate via PowerPoint lecture on Weapons Signature sources to include; a. Missiles b. Torpedoes c. Main armament (guns) (50 minutes – Method 1) 4) Instructor to explain and demonstrate via PowerPoint lecture on Decoys and acoustic Jamming sources to include; a. Narrowband / Broadband Decoys b. Narrowband / Broadband Jammers (50 minutes – Method 1) 5) Instructor to explain and demonstrate via PowerPoint lecture on Transient sources to include; a. Foghorn b. Dive noise c. Hydraulic noise d. Fluid movement e. Weapons / Decoys release f. Mast movement (Laughing Man/Babbling Brook) (50 minutes – Method 1) 6) Instructor to explain and demonstrate via PowerPoint lecture on Active Sonar sources. (50 minutes – Method 1) 7) Instructor to explain and demonstrate via PowerPoint lecture on Merchant vessel sources. (100 minutes – Method 1) 8) Instructor to explain and demonstrate via PowerPoint lecture on hydrodynamic sources to include; a. Swaths and Swath Families b. Speed Related Components (50 minutes – Method 1) 9) Students will conduct a practical lab on hydrodynamic sources. Instructor will take up the practical lab upon completion. (100 minutes – Method 3)

Instructor Resources	References
1) Sound system 2) IT support	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2

LG 001.02.02– Identify Acoustic Sources	
3) PAL Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Chronoscope-Stop watch 7) Osprey workstation- 1 per student 8) Audio Wave Files 9) PowerPoint lecture on Other Acoustic Sources	3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor	
1) Platforms are to be categorized by Hotel power.	

EO 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS
LG 001.02.03 – ANALYZING DIESEL SUBMARINES

LG 001.02.03 – Diesel Submarines		
Location Conditions		**Instructional Sequence**
Classified Classroom		After LG 001.02.02
Time	**Classification**	**Instructor**
300 minutes	Secret	AAAC Instructor
Learning Objectives		**Method(s)**
1) Diesel (SSK) Mechanics 2) Diesel (SSK) Acoustics		1) Ice Breaker 2) Lecture 3) EDI 4) Practical Lab 5) Class Debrief
Scenario	Nil	
Activity Steps	1) Instructor will conduct an Ice Breaker exercise to warm up the class in understanding Diesel (SSK) submarines. (20 minutes – Method 1) 2) Instructor will present PowerPoint lectures on Diesel (SSK) submarines to include: a. Mechanics b. Acoustics (100 minutes – Method 2 and 3) 3) Students will then complete the Diesel (SSK) submarine acoustic practical lab with a minimum of 6 contacts. (150 minutes – Method 4) 4) Instructor will conduct end of day debrief on practical lab and LG 001.03.02 Rapid Tactical Analysis of Diesel submarines (Quick Kills). (30 minutes – Method 5)	
Instructor Resources		**References**
1) Sound system 2) IT support 3) Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Osprey workstation- 1 per student 7) Diesel (SSK) submarine lab with 6 contacts		1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor		
1) Instructor(s) will be required to provide coaching, mentorship and guidance to students as required during the practical lab. 2) Instructor may use their discretion for what Ice Breaker exercise to utilize.		

EO 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS
LG 001.02.04 – ANALYZING NON-ALLIED NUCLEAR SUBMARINES

LG 001.02.04 – Non-Allied Nuclear Submarines		
Location Conditions		**Instructional Sequence**
Classified Classroom		After LG 001.02.03
Time	**Classification**	**Instructor**
300 minutes	Secret	AAAC Instructor
Learning Objectives		**Method(s)**
1) Non-Allied Nuclear Submarine Acoustics		1) Ice Breaker 2) Lecture 3) EDI 4) Practical Lab 5) Class Debrief
Scenario	Nil	
Activity Steps	1) Instructor will conduct an Ice Breaker exercise to warm up the class in understanding Non-Allied Nuclear submarines. (20 minutes – Method 1) 2) Instructor will present PowerPoint lectures on Non-Allied Nuclear submarines to include: a. Mechanics b. Acoustics (100 minutes – Method 2 and 3) 3) Students will then complete the Non-Allied Nuclear submarines acoustic practical lab with a minimum of 6 contacts. (150 minutes – Method 4) 4) Instructor will conduct end of day debrief on practical lab and LG 001.03.03 Rapid Tactical Analysis of Non-Allied Nuclear submarines (Quick Kills). (30 minutes – Method 5)	
Instructor Resources		**References**
1) Sound system 2) IT support 3) Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Osprey workstation- 1 per student 7) Non-Allied Nuclear submarine lab with 6 contacts		1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor		
1) Instructor(s) will be required to provide coaching, mentorship and guidance to students as required during the practical lab. 2) Instructor may use their discretion for what Ice Breaker exercise to utilize.		

EO 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS
LG 001.02.05 – ANALYZING ALLIED NUCLEAR SUBMARINES

LG 001.02.05 –Allied Nuclear Submarines		
Location Conditions		**Instructional Sequence**
Classified Classroom		After LG 001.02.04
Time	**Classification**	**Instructor**
350 minutes	Canadian Eyes Only Secret	AAAC Instructor

Learning Objectives	**Method(s)**
1) Allied Nuclear Submarine Acoustics	1) Ice Breaker 2) Lecture 3) EDI 4) Practical Lab 5) Class Debrief

Scenario	Nil
Activity Steps	**The following lesson is CANADIAN EYES ONLY (CEO), no Foreign Nationals will receive this instruction, practical or debrief.** 1) Instructor will conduct an Ice Breaker exercise to warm up the class in understanding Allied Nuclear submarines. (20 minutes – Method 1) 2) Instructor will present PowerPoint lectures on Allied Nuclear submarines acoustics. (100 minutes – Method 2 and 3) 3) Students will then complete the Allied Nuclear submarines acoustics practical lab with a minimum of 8 contacts. (200 minutes – Method 4) 4) Instructor will conduct end of day debrief on practical lab and LG 001.03.04 Rapid Tactical Analysis of Allied Nuclear submarines (Quick Kills). (30 minutes – Method 5)

Instructor Resources	**References**
1) Sound system 2) IT support 3) Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Osprey workstation- 1 per student 7) Allied Nuclear submarine lab with 8 contacts	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) CFCD 113 Vol 2 CEO Annex 4) National Acoustic Library 5) OSPREY Operator Manual

Note(s) to Instructor
1) Instructor(s) will be required to provide coaching, mentorship and guidance to students as required during the practical lab. 2) Instructor may use their discretion for what Ice Breaker exercise to utilize.

EO 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS
LG 001.02.06 – ANALYZING NAVAL SURFACE VESSELS

LG 001.02.06 – Naval Surface Vessels		
Location Conditions		**Instructional Sequence**
Classified Classroom		After LG 001.02.05
Time	**Classification**	**Instructor**
300 minutes	Secret	AAAC Instructor
Learning Objectives		**Method(s)**
1) Naval Surface Vessels Mechanics 2) Naval Surface Vessels Acoustics		1) Ice Breaker 2) Lecture 3) EDI 4) Practical Lab 5) Class Debrief
Scenario	Nil	
Activity Steps	1) Instructor will conduct an Ice Breaker exercise to warm up the class in understanding Naval Surface Vessels. (20 minutes – Method 1) 2) Instructor will present PowerPoint lectures on Naval Surface Vessels to include: a. Mechanics b. Acoustics (50 minutes – Method 2 and 3) 3) Students will then complete the Naval Surface Vessels acoustic practical lab with a minimum of 6 contacts. (200 minutes – Method 4) 4) Instructor will conduct end of day debrief on practical lab and Quick Kills. (30 minutes – Method 5)	
Instructor Resources		**References**
1) Sound system 2) IT support 3) Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Naval Surface Vessels lab with 6 contacts 7) Osprey workstation- 1 per student		1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor		
1) Instructor(s) will be required to provide coaching, mentorship and guidance to students as required during the practical lab. 2) Instructor may use their discretion for what Ice Breaker exercise to utilize.		

EO 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS
LG 001.02.07 – CLASSIFY PASSIVE ACOUSTIC CONTACTS

LG 001.02.07 – Classify Passive Acoustic Contacts		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After LG 001.02.06	
Time	**Classification**	**Instructor**
2600 Minutes	Secret	AAAC Instructor
Learning Objectives	**Method(s)**	
1) Contact correlation 2) Acoustic source identification 3) Passive acoustic contact classification 4) Acoustic intelligence data compilation	1) Ice Breaker 2) Practical labs 3) Class Debrief	
Scenario	Students will complete a total of 10 mixed labs over 7 days.	
Activity Steps	**Day #1** 1) Instructor will conduct an Ice Breaker exercise to warm up the class in preparation for Mixed Lab #1. (20 minutes – Method 1) 2) Instructor will assign Mixed Lab #1, students will analyze and classify the contacts. Instructor will conduct debrief on Mixed Lab #1. (330 minutes – Method 2 and 3) **Day #2** 3) Instructor will conduct an Ice Breaker exercise to warm up the class in preparation for Mixed Lab #2. (20 minutes – Method 1) 4) Instructor will assign Mixed Lab #2, students will analyze and classify the contacts. Instructor will conduct debrief on Mixed Lab #2. (330 minutes – Method 2 and 3) **Day #3** 5) Instructor will conduct an Ice Breaker exercise to warm up the class in preparation for Mixed Lab #3. (20 minutes – Method 1) 6) Instructor will assign Mixed Lab #3, students will analyze and classify the contacts. Instructor will conduct debrief on Mixed Lab #3. (330 minutes – Method 2 and 3) **Day #4** 7) Instructor will conduct an Ice Breaker exercise to warm up the class in preparation for Mixed Lab #4. (20 minutes – Method 1)	

LG 001.02.07 – Classify Passive Acoustic Contacts
8) Instructor will assign Mixed Lab #4, students will analyze and classify the contacts. Instructor will conduct debrief on Mixed Lab #4. (330 minutes – Method 2 and 3) **Day #5** 9) Instructor will conduct an Ice Breaker exercise to warm up the class in preparation for Mixed Labs #5 and 6. (20 minutes – Method 1) 10) Instructor will assign Mixed Lab #5, students will analyze and classify the contacts. (180 minutes – Method 2) 11) Instructor will assign Mixed Lab #6, students will analyze and classify the contacts. (180 minutes – Method 2) 12) Instructor will conduct a Class Debrief on Mixed Labs # 5 and 6. (20 minutes – Method 3) **Day #6** 13) Instructor will conduct an Ice Breaker exercise to warm up the class in preparation for Mixed Labs #7 and 8. (20 minutes – Method 1) 14) Instructor will assign Mixed Lab #7, students will analyze and classify the contacts. (180 minutes – Method 2) 15) Instructor will assign Mixed Lab #8, students will analyze and classify the contacts. (180 minutes – Method 2) 16) Instructor will conduct a Class Debrief on Mixed Labs # 7 and 8. (20 minutes – Method 3) **Day #7** 17) Instructor will conduct an Ice Breaker exercise to warm up the class in preparation for Mixed Labs #9 and 10. (20 minutes – Method 1) 18) Instructor will assign Mixed Lab #9, students will analyze and classify the contacts. (180 minutes – Method 2)

LG 001.02.07 – Classify Passive Acoustic Contacts	
19) Instructor will assign Mixed Lab #10, students will analyze and classify the contacts. (180 minutes – Method 2) 20) Instructor will conduct a Class Debrief on Mixed Labs # 9 and 10. (20 minutes – Method 3) 21) Instructor will conduct a Class Review prior to PC 00. (20 minutes – Method 3)	
Instructor Resources	**References**
1) Sound system 2) IT support 3) Acoustic Response Sheet 4) Projector / Monitor 5) Headsets 6) Chronoscope/Stop watch 7) Osprey workstation- 1 per student 8) 10 x Mixed Labs (minimum of 6 contacts)	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor	
1) Instructor(s) to progress class with less mentorship assistance with aim towards achieving 20 minutes per contact (test conditions). 2) Instructor may use their discretion for what Ice Breaker exercise to utilize. 3) Final 2 labs to be treated as test conditions.	

EO 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS
EC 001.02 – CONDUCTING ACOUSTIC GRAM ANALYSIS

EC 001.02 – Conducting Acoustic Gram Analysis		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.02	Classroom	After LG 001.02.07
Time	**Classification**	**Instructor**
300 minutes	Secret	AAAC Instructor
Scenario	At Instructor(s) discretion	
Assessment Steps	**Input** – Student is given a mixed lab with 6 contacts. **Performance** – Student will conduct full Passive Acoustic analysis on the contacts provided, fill out the ARS and provide contact classification. (200 minutes) **Output** – Trainee must attain 75% of both analysis and classification to pass this assessment. **Feedback** –Student is provided written feedback via the ARS. Instructor conducts class debrief. (100 minutes)	
Instructor Resources		
1) Acoustic Response Sheet 2) Projector / Monitor 3) Headsets 4) Chronoscope-Stop watch 5) Osprey workstation- 1 per student 6) Audio Wave Files 7) EC Mixed Lab		
Note(s) to Instructor		
1) EC 001.02 is a Critical EC. The student must achieve the standard IOT challenge the PC. 2) EC 001.02 will contain 6 mixed contacts covering all aspects of material taught. 3) The pass mark for this EC is 75% for analysis and 75% for classification. 4) This EC represents 20% of the overall course mark. 5) The mark to be included in the overall course mark will be determined by weighting this test result 80% for Analysis and 20% for Classification. 6) References will be permitted.		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this EC will result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this EC, continuation of training without a re-write may be recommended. See Chapter 3: Assessment of Trainee for further details.	

Acoustic Response Sheet Example

UNCLASSIFIED (SECRET WHEN COMPLETE)

Contact # _____ Date Analyzed: _____ Analyzed By: _____ Measurement Time: _____

Tactical Data

CPA Time	Dop Spd	Range	Depth
1 (± 30 s)	1 (± 2kts)	1 (± 50%)	1 (± 50%)
	TPK Spd	TPK	
	1 (± 1 kt)	1	

Aspect	Bow ADR	Bow Freq	Stern ADR	Stern Freq
1				

Propulsion

Blade Rate	# Blades	Shaft Rate	SRPM	Turb	Pred Harm	CFR	CSR	EFR	ERPM	Pred Harm
1 (± 0.5)	1	1 (± 0.2)	1 (± 12)	1 (± 0.2)	1 pt each for turbine harmonics blade harmonics shaft harmics (½ to be listed)	1 (± 0.1)	1 (± 0.2)	1 (± 0.5)	1 (± 12)	In order of predom. 1 (½ to be listed)

Auxiliaries Augmenters Unknowns

Source	Fund	Harm
Auxiliaries 1 pt for the 1st Aux 1 pt for every 3rd thereafter ie 1 listed = 1pt 4 listed = 2 pts 7 listed = 3 pts 10 listed = 4 pts	(± 0.2)	
Significant Sources 1 pt each	(± 0.2)	
Unknowns no value		
Class ID/Indicator 1	1 (± 0.2)	1 (½ to be listed) if applic.)

Swaths

Fund	Bandwidth
1 pt for the 1st 1 pt for every 3rd thereafter ie 1 listed = 1pt 4 listed = 2 pts 7 listed = 3 pts 10 listed = 4 pts	Freq tolerance (± 2%)
Class ID/ Indicator 2 pt each (1 for name 1 for freq)	Freq tolerance (± 2%)
Harmonic PRS 2 pt each (1 for freq 1 for Harmonics)	

Drive Mode	Reduction Ratio
1	1

Shaft and Engine Related Sources

Source	Fund	Harm
1	**Class ID/Indicator** 1 (± 0.5)	1 (½ to be listed) (if applicable)
1	**Modulation** 1 (± 0.5)	1 (½ to be listed)
	Other 1 (± 0.5)	

SRCs

Fund	Characteristics
1 (± 0.5)	1 (for listing the name)

Secondary Contacts

BR	# Blades	SR	SRPM	Pred Harm	TPK	TPK Speed

Transient, Significant Events

Time	Event/Freq/Harm
	1 pt for each significant entry

Aural Characteristics

	1 pt for each significant entry

Classification

4 total - broken down as 1 each for Surf/Subsurf, Naval/Nuclear/ Diesel, Type, Class

Justification	Analysis Summary
1 pt ## the info listed justifies classification	

Analysis Mark **Classification Mark**

UNCLASSIFIED (SECRET WHEN COMPLETE)

EO 001.03 – CONDUCT TACTICAL ANALYSIS
LG 001.03.01 – INTERPRET TACTICAL EVENTS

LG 001.03.01 – Interpret Tactical Events		
Location Conditions	**Instructional Sequence**	
Classified Classroom	After EC 001.02	
Time	**Classification**	**Instructor**
100 minutes	Secret	AAAC Instructor
Learning Objectives	**Instructional Method(s)**	
1) Dynamic events 2) Tactical implications of passive acoustic data 3) Changes in contact readiness/ tactical states	1) Lecture 2) Demo	
Scenario		
Activity Steps	1) Instructor will give lecture via Power Point showing examples of extracting Tactical events and demonstrating how to identify and interpret the following; a. Dynamic events to include: i. Turns ii. Speed change iii. Depth change b. Tactical implications of passive acoustic data to include: i. Aspect ii. Interference patterns to include; 1. Lloyds mirror 2. Surface patterns iii. Fade in/ out iv. Abrupt in/out c. Changes in contact readiness/ tactical states to include: i. Knee in / out ii. Intensity iii. Change in cavitation iv. Transients (100 minutes – Method 1&2)	
Instructor Resources	**References**	
1) Sound system 2) IT support 3) Projector / Monitor 4) Headsets 5) Osprey workstation- 1 per student 6) Audio Wave Files 7) PowerPoint lecture on Tactical events with examples	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual	
Note(s) to Instructor		
Nil		

EO 001.03 – CONDUCT TACTICAL ANALYSIS
LG 001.03.02 –TACTICAL ANALYSIS OF DIESEL SUBMARINES

LG 001.03.02 –Tactical Analysis of Diesel Submarines		
Location Conditions		**Instructional Sequence**
Classified Classroom		During LG 001.02.03 after Activity #2
Time	**Classification**	**Instructor**
50 minutes	Secret	AAAC Instructor
Learning Objectives		**Method(s)**
1) Rapid Tactical analysis of Diesel Submarines		1) Practical Lab
Scenario	Nil.	
Activity Steps	1) Instructor will lead a practical lab on the Rapid Tactical Analysis (Quick Kills) on Diesel submarines with 10 contacts. This will be conducted during LG 001.02.03 after Activity #2. 2) Instructor will provide students with diesel submarine contacts, students will be required to analyze and classify with a time limit of 5 minutes per contact. (50 minutes –5 minutes per contact, Method 1)	
Instructor Resources		**References**
1) Sound system 2) IT support 3) Projector / Monitor 4) Headsets 5) Osprey workstation- 1 per student 6) Diesel submarine Quick Kill lab with 10 contacts 7) Quick Kill ARS		1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor		
1) Nil		

EO 001.03 – CONDUCT TACTICAL ANALYSIS
LG 001.03.03 –TACTICAL ANALYSIS OF NON-ALLIED NUCLEAR SUBMARINES

LG 001.03.03 –Tactical Analysis of Non-Allied Nuclear Submarines		
Location Conditions	**Instructional Sequence**	
Classified Classroom	During LG 001.02.04 after Activity #2	
Time	**Classification**	**Instructor**
50 minutes	Secret	AAAC Instructor
Learning Objectives	**Method(s)**	
1) Rapid Tactical analysis of Non-Allied Nuclear Submarines	1) Practical Lab	
Scenario	Nil.	
Activity Steps	1) Instructor will lead a practical lab on the Rapid Tactical Analysis (Quick Kills) on Non-Allied Nuclear submarines with 10 contacts. This will be conducted during LG 001.02.04 after Activity #2. 2) Instructor will provide students with non-allied nuclear submarine contacts, students will be required to analyze and classify with a time limit of 5 minutes per contact. (50 minutes –5 minutes per contact, Method 1)	
Instructor Resources	**References**	
1) Sound system 2) IT support 3) Projector / Monitor 4) Headsets 5) Osprey workstation- 1 per student 6) Non-Allied Nuclear submarine Quick Kill lab with 10 contacts 7) Quick Kill ARS	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual	
Note(s) to Instructor		
1) Nil		

EO 001.03 – CONDUCT TACTICAL ANALYSIS
LG 001.03.04 –TACTICAL ANALYSIS OF ALLIED NUCLEAR SUBMARINES

LG 001.03.04 –Tactical Analysis of Allied Nuclear Submarines		
Location Conditions		**Instructional Sequence**
Classified Classroom		During LG 001.02.05 after Activity #2
Time	**Classification**	**Instructor**
50 minutes	Canadian Eyes Only Secret	AAAC Instructor
Learning Objectives		**Method(s)**
1) Rapid Tactical analysis of Allied Nuclear Submarines		1) Practical Lab
Scenario	Nil.	
Activity Steps	**The following lesson is CANADIAN EYES ONLY (CEO), no Foreign Nationals will receive this instruction, practical or debrief.** 1) Instructor will lead a practical lab on the Rapid Tactical Analysis (Quick Kills) on Allied Nuclear submarines with 10 contacts. This will be conducted during LG 001.02.05 after Activity #2. 2) Instructor will provide students with allied nuclear submarine contacts, students will be required to analyze and classify with a time limit of 5 minutes per contact. (50 minutes –5 minutes per contact, Method 1)	
Instructor Resources		**References**
1) Sound system 2) IT support 3) Projector / Monitor 4) Headsets 5) Osprey workstation- 1 per student 6) Allied Nuclear submarine Quick Kill lab with 10 contacts 7) Quick Kill ARS		1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) CFCD 113 Vol 2 CEO Annex 4) National Acoustic Library 5) OSPREY Operator Manual
Note(s) to Instructor		
1) Nil		

EO 001.03 – CONDUCT TACTICAL ANALYSIS
LG 001.03.05 –TACTICAL ANALYSIS OF NAVAL SURFACE VESSELS

LG 001.03.05 –Tactical Analysis of Naval Surface Vessels		
Location Conditions		**Instructional Sequence**
Classified Classroom		During LG 001.02.06 after Activity #2
Time	**Classification**	**Instructor**
50 minutes	Secret	AAAC Instructor
Learning Objectives		**Method(s)**
1) Rapid Tactical analysis of Naval Surface Vessels		1) Practical Lab
Scenario	Nil.	
Activity Steps	1) Instructor will lead a practical lab on the Rapid Tactical Analysis (Quick Kills) on Naval Surface Vessels with 10 contacts. This will be conducted during LG 001.02.06 after Activity #2. 2) Instructor will provide students with naval surface contacts, students will be required to analyze and classify with a time limit of 5 minutes per contact. (50 minutes –5 minutes per contact, Method 1)	
Instructor Resources		**References**
1) Sound system 2) IT support 3) Projector / Monitor 4) Headsets 5) Osprey workstation- 1 per student 6) Naval Surface Vessels Quick Kill lab with 10 contacts 7) Quick Kill ARS		1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual
Note(s) to Instructor		
1) Nil		

EO 001.03 – CONDUCT TACTICAL ANALYSIS
LG 001.03.06 – RAPID TACTICAL ANALYSIS

LG 001.03.06 – Rapid Tactical Analysis		
Location Conditions	**Instructional Sequence**	
Classified Classroom	During LG 001.02.07	
Time	**Classification**	**Instructor**
150 minutes	Secret	AAAC Instructor
Learning Objectives	**Method(s)**	
1) Rapid Tactical analysis of Mixed Contacts	1) Practical Lab	
Scenario	Nil.	
Activity Steps	1) Instructor will lead a practical lab on the Rapid Tactical Analysis (Quick Kills) on mixed contacts labs with 10 contacts. 2) Instructor will provide students with a Mixed contact lab #1, students will be required to analyze and classify with a time limit of 5 minutes per contact. This Quick Kill lab will be conducted during LG 001.02.07 Day 1 after Activity #1. (50 minutes –5 minutes per contact, Method 1) 3) Instructor will provide students with a Mixed contact lab #2, students will be required to analyze and classify with a time limit of 5 minutes per contact. This Quick Kill lab will be conducted during LG 001.02.07 Day 2 after Activity #3. (50 minutes –5 minutes per contact, Method 1) 4) Instructor will provide students with a Mixed contact lab #2, students will be required to analyze and classify with a time limit of 5 minutes per contact. This Quick Kill lab will be conducted during LG 001.02.07 Day 3 after Activity #5. (50 minutes –5 minutes per contact, Method 1)	
Instructor Resources	**References**	
1) Sound system 2) IT support 3) Projector / Monitor 4) Headsets 5) Osprey workstation- 1 per student 6) 4 x Mixed contact labs with 10 contacts per lab 7) Quick Kill ARS	1) CFCD 113 Vol 1 2) CFCD 113 Vol 2 3) National Acoustic Library 4) OSPREY Operator Manual	
Note(s) to Instructor		
1) Nil		

EO 001.03 – CONDUCT TACTICAL ANALYSIS
EC 001.03 – CONDUCT TACTICAL ANALYSIS

EC 001.03 – Conduct Tactical Analysis		
Scope	**Location Conditions**	**Instructional Sequence**
EO 001.03	Classified Classroom	After LG 001.03.06
Time	**Classification**	**Instructor**
100 minutes	Secret	AAAC Instructor
Scenario	Nil.	
Assessment Steps	**Input** – Student is given a mixed lab with 10 contacts. **Performance** – Student will conduct Tactical Analysis on the contacts provided, fill out the ARS and provide contact classification. (50 minutes) **Output** – Trainee must attain 75% to pass this assessment. **Feedback** –Student is provided written feedback via the ARS. Instructor conducts class debrief. Students may discuss answers with instructor. Instructor will have 1 hour to grade ARS. (50 minutes for debrief)	
Instructor Resources		
1) EC 001.03 will be a practical assessment based on content from EO 001.03. 2) This test consists of 10 contacts that represent a cross section of the material instructed. Students are required to complete an ARS for each contact. 3) Pass mark will be 75%. 4) This EC represents 20% of the overall course mark. 5) References are permitted.		
Note(s) to Instructor		
1) Nil		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this EC will result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this EC, continuation of training without a re-write may be recommended. See Chapter 3: Assessment of Trainee for further details.	

Quick Kill Acoustic Response Sheet Example

Contact #	CPA Time	Spd	Range	Course	Aspect	Depth	SRPM	Mode
QK #1	1	1	1	1	1	1	1	1

Aural/Transients	Tactical Info	Secondary Contacts	Classification/Justification
1	1	1	2
			1

PO 001 – ANALYZE ACOUSTIC DATA
PC 001 – ANALYZE ACOUSTIC DATA

PC 001 – Analyze Acoustic Data		
Scope	**Location Conditions**	**Instructional Sequence**
PO 001	Classified Classroom	After EC 001.03
Time	**Classification**	**Instructor**
300 minutes	Secret	AAAC Instructor
Scenario	Nil.	
Assessment Steps	**Input** – Student will be given a mixed lab with 6 contacts. **Performance** – Student will conduct full Passive Acoustic analysis on the contacts provided, fill out the ARS and provide contact classification. (200 minutes) **Output** – Trainee must attain 75% of both analysis and classification to pass this assessment. **Feedback** –Student is provided written feedback via the ARS. Instructor conducts class debrief. Students may discuss answers with instructor. Instructor will have 2 hour to grade ARS. (100 minutes)	
Instructor Resources		
1) Projector / Monitor 2) Headsets 3) Chronoscope-Stop watch 4) Osprey workstation- 1 per student 5) Audio Wave Files 6) Acoustic Response Sheet 7) PC Mixed Lab with 6 contacts		
Note(s) to Instructor		
1) The student must achieve the standard IOT pass the course. 2) PC 001 will contain 6 mixed contacts covering all aspects of material taught in EO 001.01, EO 001.02 and EO 001.03. 3) The pass mark for this EC is 75% for both analysis and for classification. 4) The mark to be included in the overall course mark will be determined by weighting this test result 80% for Analysis and 20% for Classification. 5) References will be permitted.		
Assessment Limitations	Nil.	
Impact of Failure	Failure to achieve the standard on this PC will result in a retest and/or PRB, IAW NPTORD 4500-54. Upon Independent Review (IR) as a result of failure of this PC, continuation of training without a re-write may be recommended. See Chapter 3: Assessment of Trainee for further details.	

Acoustic Response Sheet Example

UNCLASSIFIED (SECRET WHEN COMPLETE)

Contact # _____ Date Analyzed: _____ Analyzed By: _____ Measurement Time: _____

Tactical Data

CPA Time	Dop Spd	Range	Depth
1 (± 30 s)	1 (± 2kts)	1 (± 50%)	1 (± 50%)
	TPK Spd	TPK	
	1 (± 1 kt)	1	

Aspect	Bow ADR	Bow Freq	Stern ADR	Stern Freq
1				

Propulsion

Blade Rate	# Blades	Shaft Rate	SRPM	Turb	Pred Harm	CFR	CSR	EFR	ERPM	Pred Harm
1 (± 0.5)	1	1 (± 0.2)	1 (± 12)	1 (± 0.2)	1 pt each for turbine harmonics blade harmonics shaft harmics (½ to be listed)	1 (± 0.1)	1 (± 0.2)	1 (± 0.5)	1 (± 12)	In order of predom. 1 (½ to be listed)

Auxiliaries-Augmenters-Unknowns

Source	Fund	Harm
Auxiliaries 1 pt for the 1st Aux 1 pt for every 3rd thereafter ie 1 listed = 1pt 4 listed = 2 pts 7 listed = 3 pts 10 listed = 4 pts	(± 0.2)	
Significant Sources 1 pt each	(± 0.2)	
Unknowns no value		
Class ID/Indicator 1	1 (± 0.2)	1 (½ to be listed) if applic.)

Swaths

Fund	Bandwidth
1 pt for the 1st 1 pt for every 3rd thereafter ie 1 listed = 1pt 4 listed = 2 pts 7 listed = 3 pts 10 listed = 4 pts	Freq tolerance (± 2%)
Class ID/ Indicator 2 pt each (1 for name 1 for freq)	Freq tolerance (± 2%)
Harmonic PRS 2 pt each (1 for freq 1 for Harmonics)	

Drive Mode	Reduction Ratio
1	1

Shaft and Engine Related Sources

Source	Fund	Harm
1	**Class ID/Indicator** 1 (± 0.5)	1 (½ to be listed) (if applicable)
1	**Modulation** 1 (± 0.5)	1 (½ to be listed)
	Other 1 (± 0.5)	

SRCs

Fund	Characteristics
1 (± 0.5)	1 (for listing the name)

Secondary Contacts

BR	# Blades	SR	SRPM	Pred Harm	TPK	TPK Speed

Transient, Significant Events

Time	Event/Freq/Harm
	1 pt for each significant entry

Aural Characteristics

	1 pt for each significant entry

Classification

4 total - broken down as 1 each for Surf/Subsurf, Naval/Nuclear/ Diesel, Type, Class

Justification	Analysis Summary
1 pt if the info listed justifies classification	

Analysis Mark _____ Classification Mark _____

UNCLASSIFIED (SECRET WHEN COMPLETE)

CHAPTER 3: ASSESSMENT OF TRAINEE

GENERAL

1. All PCs are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessments and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. All ECs are formative assessments that help the trainee and the instructors to recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Non-critical EC results do not factor into a trainee's successful completion of the PO.

3. The process for critical ECs and PCs is outlined in NPTORD 4500-54. For administrative purposes, a Critical EC has the same importance as a PC.

4. PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

5. Assessment of trainees is to be completed IAW the standards delineated in this QSP, NPTORD 4500-54, and CFITES Vol 7 – Evaluation.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Vol 6 – Conduct of Instructional Programmes.

7. Continuous monitoring of trainee progress is required to provide the following:

8. Early warning of trainee difficulties;

9. Feedback on the effectiveness of training;

10. Information for the Training Review Board (TRB) (if required); and

11. Information to assist in the completion of the CF 377 Course Report.

12. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs/SOPs and DAOD 5516-5.

13. PC failures and supplemental PC failures are to be handled IAW TE SOs/SOPs and NPTORD 4500-54. Supplemental ECs and PCs shall not be administered within 24 hours of the trainee being informed of their EC or PC failure. Trainees will have 24 hours at minimum to prepare for a retest.

TRAINING REVIEW BOARD

14. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54.

COURSE REPORTS

15. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

16. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

17. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

18. A record of trainee counselling shall be maintained for 2 years by the TE IAW Information Management Standard (IMS) 6001-2-1 and Defence Subject Classification and Disposition System (DSCDS) Primary Number 4500.

ANNEX A – REFERENCES

This table lists all references used within this QSP.

#	Short Title	Long Title
A – CAF or other Official Canadian Government Publications		
A1	CFCD 113 Vol 1	CFCD 113(E) Vol 1
A2	CFCD 113 Vol 2	CFCD 113(E) Vol 2
A3	CFCD 113(E) Vol 2 CEO Annex	CFCD 113(E) Vol 2 CEO Annex
A4	NAL	National Acoustic Library (NAL) TRINITY (ADAC)
A5	NDSOD	National Defense Security Orders and Directives (NDSODs)
A6		ADAC Acoustic Response Sheet Marking Guide
B – Official Allied Force and Allied Government Publications		
B1		
B2		
C – Commercial Publications		
C1		JANES Publications
C2		Osprey Operator Manual
D – Other Publications		

ANNEX B – TASK LIST

This table provides an overview of MTL tasks WRT if/where they are trained within this QSP. No train substantiations are located in the Record of Decisions annex.

Task #	Task Statement	T/NT	PO
T0001	Analyze acoustic data	T	PO 001
T0002	Interpret Passive Aural Listening	T	PO 001
T0003	Conduct tactical analysis	T	PO 001
T0004	Interpret dynamic events	T	PO 001
T0005	Interpret tactical implications of passive acoustic data	T	PO 001
T0006	Interpret changes in contact's readiness / tactical states	T	PO 001
T0007	Interpret acoustic sources	T	PO 001
T0008	Conduct contact correlation	T	PO 001
T0009	Identify acoustic sources	T	PO 001
T0010	Classify passive acoustic contacts	T	PO 001
T0011	Compile acoustic intelligence data	T	PO 001
Added Tasks			

ANNEX C – RECORD OF DECISIONS

The following summary provides insight to the decisions and reasoning resulting in this QSP.

Course History:

Dates: 8-16 May 2007

Location: Canadian Forces Naval Operations School (CFNOS), CFB Halifax

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Gullison	G	Capt	00182	404 Sqn
Haines	B	Lt (N)	00211	CFNOS
Stover	J	Capt	00182	405 Sqn
Sannes	R	Capt	00182	ADAC (P)
Sullivan	S	A/SLt	00207	CFNOS
Clarke	P	CPO2	00114	DMTE
Bricknell	B	PO1	00324	CFFS Esquimalt
McPhee	D	PO2	00324	ADAC (A)
Hopper	S	PO2	00324	CFNOS
Desmeules-Alain	P	MS	00324	CFNOS

Background:

1. Acoustic Data Analysis Centre (TRINITY) has delivered The Advanced Acoustic Analysis Course (AAAC) for over 4 decades. Due to changes in technology and environment, the demands and expectations of an Advanced Acoustic Analyst have evolved. There are currently several occupations within the CF that require the Advanced Acoustic Analyst specialty:

1. Acoustic Sensor Officers (ASO)in the Maritime/Long Range Patrol Community
2. Tactical Communications Officers (TACCO) in the Maritime Helicopter Community
3. Sonar Operators (SONAR OP) aboard Naval Ships and Submarines
4. Analysts and Supervisors at TRINITY Atlantic & Pacific

In response to the recognized deviations between the course and governing documentation, a QSP writing board was established to fix any problems.

Working Group Objective:

2. The mandate of the QSPWB was to:

1. Review the Advanced Acoustic Analyst OSS, and
2. Create a corresponding QSP for the Advanced Acoustic Analyst.

Specification Task – No Train Decisions:
Annex B contains all tasks associated with Advanced Acoustic Analyst OSS. All tasks shall be trained.

Task #	Task Statement	Substantiation

Minutes:
3.
Day 1:

0930 – CO ADAC (A) briefed board on history of the course and expectations.
1015 – Board TDO Lt (N) Blair briefed the board on QSP process and introduced A/SLt Sullivan. She would be providing assistance to the board as part of her OJE.
1030 – CPO2 Clarke briefed board on his mandate and expectations
1035 – Introduction of board members
1045 – Commenced OSS board, brainstorming on the job of Advanced Acoustic Analyst
1230 – Rough definition of the job developed
1500 – Initial Task List developed and grouped
1530 – Rough draft of OSS developed

Day 2:

0800 – QS board commenced
0926 – 1 PO & 4 EOs determined
0939 – QSP writing commenced
1130 – Rough draft of Chapter 1 and Chapter 2 of QSP completed
1300 – Instructional Analysis for the training plan commenced
1500 – Started to write EO 001.01

Day 3:

1030 – Finished writing EO 001.01
1045 – Started to write EO 001.02
1400 – Finished writing EO 001.02
1415 – Started to write EO 001.03

Day 4:

1200 – Finished EO 001.03

Day 5:

0800 – Assessment Plan commenced
1000 – EC 001.01 drafted
1400 – EC 001.02 drafted

Day 6:

0900 – EC 001.03 drafted
1100 – PC 001 drafted
1430 – Annexes completed

Day 7:

0800 – Document review conducted
1130 – Board adjourned

Day 8:

1500 - Record of proceedings drafted

Discussions / Decisions:

4. Prior to convening the board, Capt Gullison discussed with Lt (N) Haines and CO ADAC (A) potential issues associated with the OSS. Specifically, that in its current state, it does not reflect any specific job in the CF. Capt Gullison inquired whether an OSS was actually required to develop a QSP. Lt (N) Haines confirmed the requirement of an OSS to develop and conduct training.

5. During the board brief, CO ADAC (A) indicated that the OSS should not reflect any specific operational or analyst position per se. Moreover, he stated that he didn't want the course to necessarily reflect a specific level of acoustic analysis (ie. Level 3).

6. Given that an OSS for ASOs and SONAR Ops already exists, and the board had no desire to create an OSS that describes the job of a TRINITY analyst, development of an appropriate OSS would prove to be a difficult task.

7. As previously mentioned, the AAAC has consistently provided excellent training to several occupations within the Canadian Forces and this requirement still exists. Concerns were voiced from PO2 Bricknell and PO2 Hopper regarding the AAAC and its ability to meet the needs of its community. Their initial impressions were that perhaps the Sonar QL5B would be better served if they taught the advanced acoustic analysis portion of the course themselves. Given similar concerns from the ASO community, there was significant pressure on the board to ensure the creation of an OSS and corresponding QSP that meets the needs of all CF user groups.

8. It was agreed by all board members that the development of OSS based on an amalgam of the tasks performed by all the user groups would not be feasible. In particular, operationally

based tasks involving specific equipment and/or environmental conditions that differ across the trades and would not make logical sense to teach during the AAAC.

9. Due to these issues, the board decided that the job of an Advanced Acoustic Analyst would describe common tasks performed by all user groups. These tasks would be equipment and environment independent. In truth, the agreed upon tasks most closely reflect the job of a level 3 acoustic analyst.

10. Over 40 acoustic analysis tasks were taken from the old OSS, ASO QS and SONAR Op QSP. Subsequently, all environmental, operator-specific, equipment- related, and unique tasks were removed, leaving 26 tasks. These tasks were grouped into 4 main tasks and 22 sub tasks. Upon further development of the POs and EOs, the 4 main tasks were divided into the 1 PO and 3 EOs established in the QSP. Furthermore, of the remaining 22 sub tasks, 9 were retained, while the other 13 were changed to knowledge and were listed in the OSS as such.

11. There was some disagreement as to the organization of the POs and EOs. Most members initially felt that perhaps there should have been 3 POs reflecting the three main tasks: aural analysis, tactical interpretation, and gram analysis. The sub tasks would therefore be the EOs within the PO. However, after discussion with Lt (N) Haines, he convinced the board that the most efficient and effective organization of the job was to turn the 3 POs into EOs, and the EOs into teaching points. As the course is only 3 three weeks long, 1 PO would suffice. In fact, this methodology reduced the workload on the board significantly without hindering the outcome.

12. As mentioned, the job was divided into three main tasks: aural analysis, tactical interpretation, and gram analysis. This job definition jumpstarted conversation on the potential for the course. Currently, the bulk of the AAAC is geared to gram analysis. Tactical interpretation and aural analysis (PAL) were treated as subtasks of gram analysis, and therefore, were not trained in significant detail. Critique of the AAAC outlined this shortfall. Given this new definition of the job, however, PAL and tactical analysis are treated independently and will be given additional weight. Although it was argued successfully by Capt Sannes that PAL and tactical analysis are, in fact, subcomponents of gram analysis, the board saw the benefits of separating them into independent EOs. It is expected that these changes will please the ASO and SONAR Op trades.

13. One of the questions that did arise from the development of the OSS was the level to which these tasks had to be performed. The previous OSS stated that the tasks and knowledge were to be performed at level 3 (indicating detailed knowledge and competent skill). Intuitively, this created a conflict for the ASO community. ASOs get trained to level 3 knowledge and skill during MOAT. If there were no delta training between MOAT and AAAC, there would be no need for ASOs to attend AAAC. In order for ASOs to upgrade from B category to a category, they need to demonstrate level 4 knowledge. Ideally, AAAC would provide that training.

14. As the board discussed this issue, differing opinions. First, level 4 knowledge requires extensive experience through on job performance. It was argued by Lt (N) Haines that this level of training couldn't be provided through formal training. The counter argument by Capt Gullison

postulated that simulation and the provision of an appropriate number of training contacts might provide the enhanced knowledge required.

15. Secondly, it was argued by the board that level 3 knowledge attributed to an Advanced Acoustic Analyst does not equate to level 3 knowledge of an ASO. Due to the tasks associated with an Advanced Acoustic Analyst, this would require more extensive knowledge, specifically in PAL, ship and submarine mechanics, acoustic sources, and source characteristics. These teaching points would certainly aid in developing an ASO's enhanced acoustic knowledge.

16. There was some concern expressed by Capt Gullison with respect to QSP development and the specific nuances that Navy IT&E have applied to CFITES. Specifically, there was apprehension that Navy OSS/QSP no longer describes skills as a subcomponent of tasks. Under current policy, tasks are supported by sub-tasks and knowledge-based teaching points. This lends to a tendency, as noted in the board, to reduce teaching points (TPs) to knowledge. The result of this process leads to increased classroom instruction, which potentially defies the principle of performance- oriented instruction.

17. Specifically, as no tasks (or skills) were listed in EO 001.01, there was a potential for this EO to be taught entirely through classroom lectures and tested through a quiz. However, the subject matter experts (SME) recognized that PAL is a valuable skill that requires practice to develop. Therefore, exercises were inserted into the training plan to ensure that students acquire PAL skills. The assessment of this EO was also a skill- based exercise.

18. A significant portion of the board was dedicated to determining an appropriate assessment plan. The format of the current AAAC has two proficiency assessments: an EC in the middle of the course and a PC on the second to last day. Capt Gullison expressed concern that without conducting enough quantitative assessments during the course, the potential for subsequent failure increases significantly. Furthermore, he argued that due to the nature of the assessments, too much time was attributed to administration. He suggested that perhaps all labs should be assessed so that a running tally of performance could be noted.

19. Capt Sannes and PO2 McPhee were concerned that assessing each lab would require extensive resources that were not available. Capt Gullison replied that students could mark labs by exchanging answer sheets amongst students. This would allow for simultaneous debrief and marking. MS Desmeules-Alain argued that only appropriate standards personnel are qualified to mark exams and that this informal marking method would incite errors and problems. Capt Gullison reiterated his desire to institute an informal assessment methodology (assess each assignment and count the marks to a final grade) to allow students to progress through the course without feeling anxiety over a few important tests.

20. After all discussion, it was decided to have 3 ECs: one for PAL, one for tactical analysis, and one for gram analysis. Furthermore, the EC for gram analysis would be broken into three parts (as was done in the past), and would be a critical EC.

21. With this discussion, the pass mark, weighting of the ECs, and the final grade were established. The pass mark for each EC and the course is 75%. The student must achieve 75% in

both analysis and classification. PAL would contribute 10% to the final; tactical analysis 20%; and gram analysis 70% (of which 80% would be analysis, 20% classification). Lastly, the average of the best two of three-gram analysis results would count toward the final, allowing for a 'bad day'. It was agreed that although this was a convoluted assessment plan, it represented the requirements well.

Guidance to Instructional Designers:

22. In the development of the training plan, issues of sequencing came up several times. There was concern that the teaching points associated with gram analysis and tactical analysis was co-dependent and could not be taught linearly. For example, IOT conduct gram analysis students would need to know tactical analysis. However, tactical analysis is a skill that requires a basis in gram analysis. Therefore, instructional designers will have to teach parts of EO 001.03 while teaching EO 001.02.

23. Secondly, under the current AAAC schedule, after students complete the gram analysis exam, the entire afternoon is used for marking. Capt Gullison considered this valuable time that was being wasted. Therefore, creative scheduling will be required to ensure that students conduct the exams in the morning and still have appropriate classes scheduled for the afternoon. Suggestions brought forward were tactical analysis lectures and exercises.

24. The board provided a framework for content and assessment. The exact details and specifics will need to be determined in the development phase.

Follow-up Action:

25. This QSP should be reviewed upon completion of the pilot course.

26. ADAC (A) will need to develop training related documents and orders to govern the conduct of AAAC to include progress monitoring and review.

Item	Action	OPI	Expected Date of Completion

Dates: 18 June 2008

QSP Amendment S & P Review Version 2.0

1. Updates were based on the 10 April 2008 Pilot serial After Action Report by ADAC (A) Trg O, Capt Gullison. Relevant portions of that report are excerpted here.

ADVANCED ACOUSTIC ANALYSIS PILOT COURSE

2. In May 2007, an OSS/QSP writing board was held to re-examine the Advanced Acoustic Analysis Qualification (AEJR) and to create the requisite training documentation to properly

conduct subject training, ref A. Subsequently, IAW ref B, ADAC (A) was to affect changes to the AAAC in order to meet the requirements of the QSP, conduct a pilot course in February 2008, and report to DMTE on the pilot course NLT 18 Apr 08.

EO 001.01 PASSIVE AURAL LISTENING

3. As indicated, much of the courseware for this objective was derived from existing training materials at ADAC (A). However, considerable effort was required to tailor the materials to fit the specific requirements of the QSP. Specifically, exercises and exams had to be created, and significant portions of the training had to be compressed due to the constraints of a three-week schedule. This was proven extremely difficult, as the QSP expected a significant level of proficiency in PAL, yet only 11 periods were provided to teach, exercise and test students on the essential skill.

4. The teaching points in the QSP are generally very appropriate and a basic introduction could be completed in the time allotted. However, in order to acquire the skill, students needed significantly more exposure, practice and testing. In retrospect, given that no submariner was part of the QSPWB, the board grossly underestimated the amount of time required to train the students in PAL.

5. From the course critique of the pilot course, there was obvious enthusiasm from the students. They thoroughly enjoyed the material and found it extremely worthwhile, however, they all indicated that the pace was too frenetic for adequate learning to occur. Moreover, exam results were indicative of this viewpoint. Student's average scores were in the mid 50% range, while the pass mark is 75%. Therefore, it is recommended that EO 001.01 be increased to 33 periods. Although some reductions were made and efficiencies were gained in other areas of the course, it is not possible to conduct the AAAC in 15 training days as directed.

EO 001.02 Detailed Analysis

6. Given that the previous version of the AAAC singularly taught detailed analysis, most of the courseware was already developed for this objective. However, as the new QSP correctly insists on two additional disciplines (PAL and Tactical Analysis), efficiencies had to be gained in this EO in order to meet the course schedule.

7. Specifically, analysis exercises were reduced from 6 to 5 contacts and from 5 to 3 hours each. Furthermore, the associated lessons were also compressed from 2 hours to 1 hour each. Overall, this compression had an impact on final test scores, which averaged in the low to mid 80% range. As a result, it is recommended that the lessons revert back to 2 hours each and that one of the conventional diesel exercises be changed to an additional mixed lab. With improved instruction on submarine mechanics and acoustic sources and the additional mixed lab, it is likely that scores will increase substantially.

8. Another issue hampering this EO is quality control of the training contacts. Given the requirement for students to quickly and accurately analyze each and every source in the LOFAR gram, it is essential that every contact be analyzed to the highest level possible. This is only

accomplished if each training contact undergoes an extensive quality assurance (QA) process in the ADAC (A) Operations department. Currently there are 125 training contacts used on the course. During the pilot course, less than 25% of the course contacts had completed a full QA. This resulted in questions from students on the accuracy and validity of training, as viewed in the critique, significantly hindering student learning and feedback. This will continue to be a challenge for ADAC (A) over the next few serials; however, it continues to be a firm requirement for EO 001.02. As each contact is fully QAd, the AAAC will show continual improvement.

9. Testing is another concern with this EO. As it is the focal point for advanced analysis training, the enabling check (EC) is considered critical for the evaluation of students. Given the unique characteristics of acoustic analysis, often students will incorrectly analyze an entire contact, yet that is not necessarily a true indication of their analysis skills. Therefore, it is essential to give students enough contacts to demonstrate that skill. This was accomplished on the pilot course by allowing students to conduct three analysis exams and discarding their worst result (best 2 of 3). Although a very valid and reliable testing technique, it required too much time in the schedule to conduct and allowed for students to forego an entire test if they succeeded on the first two.

10. Therefore, to achieve the requisite test reliability and balancing the scheduling needs of the course, the following recommendations are made for the QSP (as indicated in Appendix 1):

a. Continue to have the students conduct two tests to ensure a valid sample and average these results;

b. Reduce the number of contacts for each test from 6 to 5;

c. Reduce the test duration from 4 to 3 hours; and

d. If students do not pass after two tests, allow them to retake one of the tests on the final day of the course.

11. Lastly, there has been a renewed focus on acoustic weapons such as torpedoes and missiles. As indicated in the QSP, there is a requirement to teach these contacts, but no exercise has been specifically attributed. As amended, ADAC (A) will include specific and focused instruction on this teaching point in the same manner as conventional diesel, nuclear and naval surface instruction.

EO 001.03 TACTICAL ANALYSIS

12. As previously discussed, specific, focused training in tactical analysis has not been traditionally conducted on the AAAC. Several course critiques identified the desire for tactical analysis training and foreign AAAC in Australia, the UK and the Netherlands currently conduct rapid analysis with great success. Although, it was determined by the QSPWB that this training was necessary, it had to be completed from base principles.

Specific teaching points and methodologies were identified by the QSP and courseware was developed to achieve the aim, however, the outcome was not as successful as envisioned.

13. There were two reasons for the shortfall. Firstly, several hours were dedicated to developing lessons on specific teaching points such as dynamic events and changes in readiness. Nearly all the students on the course were well versed in the theory; hence the lessons were unnecessarily redundant. Secondly, there were not ample examples and exercises provided to allow the students to practice their skills. As observed on foreign courses, tactical analysis is beneficial and enjoyable if conducted quickly using many examples and exercises. To summarize, it was determined that the best methodology for this EO was practice using many examples. This module will be redeveloped for Sep 08.

TIMINGS & SCHEDULING

14. In the development of the pilot course it was apparent that while some estimates for training delivery time were very accurate, many others were not. As previously discussed, PAL was significantly underestimated. However, some of the detailed and tactical analysis timings were overestimated. The QSP has been thoroughly reviewed in this regard and all amendments should be implemented.

ADMINISTRATION

15. Several administrative details need to be included to the QSP. Given that NTI 06/2007 has recently superseded NTI 02/2001, some of the guidance for the CCA is no longer captured. Specifically, there is little guidance for course scheduling. As CCA, it is recommended that ADAC (A) be given control over the scheduling of AAAC serials in order to maintain courseware IAW QSP revisions. Furthermore, attached are recommended training orders for the conduct of the AAAC (Appendix 2) to ensure quality control IAW ref D. It is recommended that they constitute Annex L of the QSP.

RECOMMENDATIONS

16. The following summarizes the recommendations for changes to the QSP. Each of these amendments can be found in Appendix 1:

a. The course needs to be lengthened to 17.4 training days;
b. PAL training needs to be increased to 33 periods;
c. Specific instruction on ASW weapons to include a lab, must be added;
d. The second diesel lab is not necessary;
e. An extra Mixed Lab is required to ensure comprehension;
f. All labs and analysis exams should consist of 5 contacts and 3 hours duration;
g. ACINT instruction needs to be reverted to 2 periods per topic from 1 hour;
h. Tactical Analysis needs to focus on exercises and examples and eliminate all theory.

Dates: 31 October 2011

QSP Amendment Version 3.0

1. Recent discussions between the TEs have brought to light discrepancies and contradictions within version 2 of this QSP. Over the course of the first six months of 2011, numerous recommendations for changes were collected and two serials of the course were run to trial these recommendations. The recommended changes and rationale are outlined below.

2. The first area of concern was the structure of the PC and ECs. The idea of a two- part evaluation, with the caveat that "If students do not pass after two tests, allow them to retake one of the tests on the final day of the course" was thought to be unsatisfactory. Representatives for both TEs, CFNOS and 404 LRP Sqn agreed at the annual Training Alignment Meeting in December 2010 that a more appropriate solution would be to have a single EC for EO001.02, and a final PC, each weighted accordingly. This would allow a more standardized approach, more in line with applicable training orders. This format was trialed on AAAC 1102B in July 2011 and was deemed to successfully address all concerns with the 2008 QSP methodology.

3. The structure of the assessment has also been re-evaluated. With the maturing of Rapid Tactical Analysis and a structured approach to analysis, students are generally able to extract relevant tactical and classification data within the first 5 minutes and are able to extract all relevant acoustic data within 30 minutes. As a result, the standard for EC 001.02 and the PC has been updated to consist of analysis of 6 contacts within 180 minutes. Using this same standard, this allows a four hour analysis lab to consist of 6 contacts, with an hour of debriefing per lab.

4. The content and weighting of EC 001.01 and 001.03 is unchanged, although there have been numerous recommendations to weight Rapid Tactical Analysis more heavily and to have it make up a greater portion of the course schedule, as it is seen by all users to more effectively suit the needs of operators. More emphasis on Tactical Analysis should be considered during the next full QSP review.

5. Students and instructors have recommended that more emphasis be placed on Diesel Submarines in battery mode. One recommendation is to change EO 001.02 to include a second Diesel Lab, concentrating on battery mode. This recommendation may necessitate a change to the duration of the course, so it should also be considered during the next full QSP review.

6. In Chapter 4, each EC and PC has a section for notes. These sections have been populated with notes on the normal consequences of failure to achieve the standard.
EC 001.01 and EC 001.03 have been treated as non-critical. Students who are unsuccessful on their first attempt may be permitted to proceed with training after debrief and/or some supplementary training without a supplementary EC. Scores will still be weighted into the final score, but merely for their effect on course standing, not on pass/fail course criteria. EC 001.02 and the PC are considered critical and must be successfully completed in order to continue on with training and to pass the course. This distinction could, in very rare cases, produce a student with an overall mark slightly below 75% who still meets course pass criteria. This risk and potential result has been deemed acceptable.

7. Classification of contacts aurally was added to the standard for PC 001 and EC 001.01, as it is currently taught and is one of the key reasons for teaching and using PAL skills.

8. Teaching Points with a time of greater than 1 period in Chapter 3 have been subdivided to more easily manage QSP compliance and course monitoring. The content of each lecture will be more easily related back to the Teaching Points covered.

9. In the Record of Discussion (Annex C) it was stated that the teaching points associated with gram analysis and tactical analysis was co-dependent and could not be taught linearly and that instructional designers will have to teach parts of EO 001.03 while teaching EO 001.02. This idea could also be expanded to include EO 001.01, as much of the PAL instruction covers concepts also dealt with in EO 001.02, such as causes of acoustic signature (TP7), and in EO 001.03, such as Changes in contact readiness and tactical states (TP3). As such, Annex G will still need updating at the next QSP Writing Board, and after trial versions of the course based on Version 3 updates have been run, to reflect optimal course sequencing with lessons grouped by topic, not by EO number.

10. Other minor amendments to correct spelling, grammar and to update administrative details such as MOS names and page numbering have been applied throughout the document.

Dates: 30 April 2019

QSP Amendment Post-Combat Training Review Version 4.0

1. Updates were based on the outcomes and recommendations of the Combat Training Review for the RQ MS / AAA Courses conducted in January 2019 by NTDC(A), NFS(A) and TRINITY ADAC staff.

RECOMMENDATIONS

2. The following summarizes the recommendations for changes to the QSP:

a. The course needs to be lengthened to 19.5 training days to accommodate TRINITY ADAC's training time requirements, RCAF training was consulted through ADAC and agreed with this recommendation;
b. PAL training needs to be decreased to 24 periods to allow for increased level of training required;
c. Addition of 1 period for lecture on how to operate OSPREY training tool, there was no current time allotted;
d. Increase in EO 001.02 to 100 periods to allow for training on new submarine platforms;
e. Update EO 001.03 to indicate the new terminology of Tactical Analysis (TIES vs. QUICK KILLS) and addition of 4 periods to provide sufficient training time.

Dates: 04 November 2020

QSP Amendment Revision RCN Rank Changes Version 4.1

IAW CANFORGEN 112/20 RCN JR Rank Designation Change 041200Z SEP 20, all previous RCN JR Rank designations in this QSP have been amended to reflect the new policy.

Dates: 08 February – 08 March 2022

QSP Amendment Revision Conversion to 3 Chapter Format Version 5.0

Location: Naval Training Development Centre (Atlantic), CFB Halifax

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Meyers	R	Lt	00211	NTDC(A)
Tallman	J.R.	CPO2	00324	NTDC(A)
Blonde	ADJ	MS	00324	NTDC(A)
Deneau	JA	MS	00324	NTDC(A)
Lavergne	S.	Civ		TRINITY ADAC
Musseau	R.	Civ		TRINITY ADAC

Background:

1. The Advanced Acoustic Analysis Course (AAAC) QSP has been in the Legacy format for many years with a separation of the QSP and Master Lesson Plan. With NPTG moving towards the new 3 Chapter QSP format with the Master Lesson Guidance incorporated into the QSP, there was a need to convert the Legacy QSP into the new format and add the Master Lesson Guidance.

In response to the need to convert the QSP, a small Working Group was established to ensure the conversion was compliant with CFITES, NPTORDs and the CCA's requirements.

Working Group Objective:

2. The mandate of the QSPWB was to:
   a. Review the Legacy Advanced Acoustic Analyst QSP,
   b. Convert the Legacy QSP to the new 3 Chapter QSP format via a contracted technical writer;
   c. Convert and/or develop all Master Lesson Guidance;
   d. Update all subsequent Annex's; and
   e. Submit the converted QSP to NPTG HQ for approval.

Specification Task – No Train Decisions:

Annex B contains all tasks associated with Advanced Acoustic Analyst OSS. All tasks shall be trained.

Task #	Task Statement	Substantiation

Minutes:

10 – 19 January 2022

- Informed NPTG HQ Training Manager (CPO2 Stewart) of intent to convert AAAC QSP to new format
- AAAC QSP reviewed by NTDC(A) CCD Analysts to determine requirement for any major and/or minor amendments
- No major amendments were identified, only requirement to convert to the new QSP format

20 January 2022

– AAAC QSP submitted to NTDC(A) Civilian contractor Susannah Daines for conversion to 3 Chapter format

26 January 2022

– Received confirmation from NPTG HQ Training Manager (CPO2 Stewart) on NTDC(A) intent to convert AAAC QSP, was recommended NTDC(A) submit DsARs to conduct conversion of QSP format

28 January 2022

– Received AAAC QSP back from technical writer converted to the new QSP format and direction on what was still required to be updated / completed

08 February 2022

- Conducted AAAC QSP Working Group with members indicated at NTDC(A) MARTLET 4
- Reviewed and updated Chapter 1 items regarding instructor qualification requirements
- Developed PO 001 EO 001.01 Master Lesson Guidance

09 February 2022

- Conducted AAAC QSP Working Group with members indicated at NTDC(A) MARTLET 4
- Developed PO 001 EO 001.02 and EO 001.03 Master Lesson Guidance
- Due to other unit requirements, remainder of QSP Working Group was placed on hold until NTDC(A) and TRINITY ADAC staff could meet on the 22 & 23 Feb 22

15 – 17 February 2022

- CPO2 Tallman reviewed QSP and made amendments in document for continuity in References, Resources, grammar and format

22 February 2022

- AAAW QSP Working Group, with the addition of TRINITY ADAC Training Manager Randy Musseau, reviewed the developed Master Lesson Guidance and made amendments to better reflect both the training content and course schedule requirements. This included adding additional time for submarine platform lectures and labs.
- **Course Capacity:**

- Amendment was made to Minimum Course Load from 6 students to 4 students. Justification for this is due to course generally being small in numbers due to SONAR Operator trade, and that course can be conducted at satellite locations such as Whidbey Island, WA, USA, a Course Alteration Request or Waiver was required to run courses below 6 students. Lowering the Minimum course load to 4 students alleviates this requirement while not negatively impacting the AAAC course.
- **Prerequisites:**
  It was determined by TRINITY ADAC Training Manager that the current Prerequisites on the DPGR Specification Details for AEJR – Advanced Acoustic Analyst were outdated based on the following criteria:

- for Officers, MPA - have completed MOAT followed by 6 months operational experience or Maritime Helicopter - be ADJJ qualified
  IAW RCAF 404 LRPA Sqn 14 Wing Greenwood Convening Order: Training Plan Review Board ASO B dated September 2020; only the AES Operator will now conduct the ASO B MOAT training and will be employed with LRPA, those with Maritime Helicopter will no longer be required to conduct the training;
- for NCMs, be QL5A qualified and have 6 months of subsequent operational experience
  QL5A is no longer used, amended to indicate the NQUAL AKYF and/or RQ-S1
- be selected for employment as an ASO in a Maritime Patrol Aircraft, SENSO in a Maritime Helicopter, staff Acoustic analyst in ADAC/DIAC/NOPF or other passive acoustic related positions
  This item is no longer required as these positions are covered under previous line items

23 February 2022

- Completion of Annex C – Record of Decisions
- Completion of Annex E – Costing Information
- Completion of Annex H Scheduling Information and Recommended Schedule
- CPO2 Tallman commenced initial QSP review, upon completion will submit to TRINITY ADAC Randy Musseau (Trg Manager) for their review.
- NTDC(A) review completed and submitted to TRINITY ADAC Training Manager for review

08 March 2022

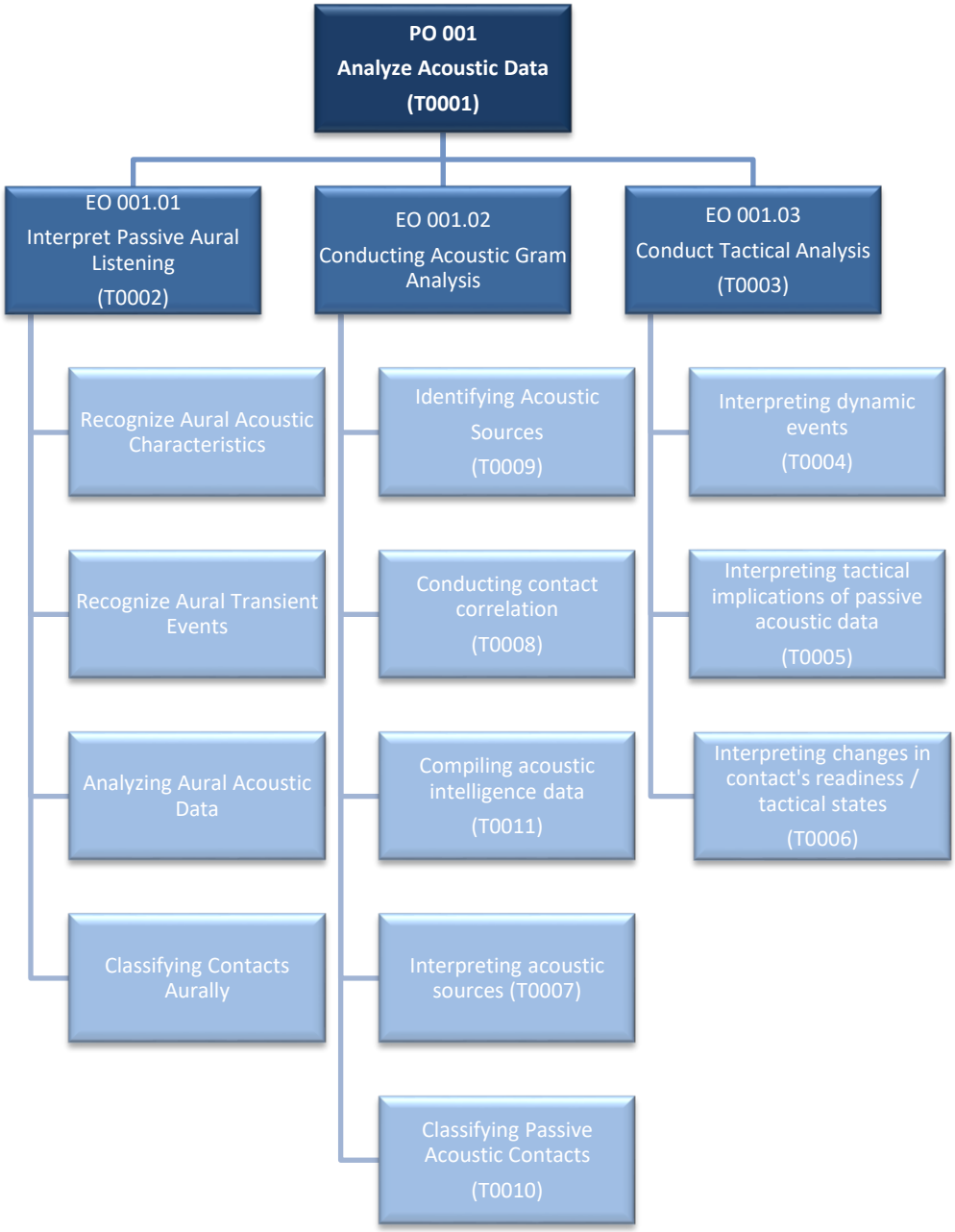
- TRINITY ADAC completed review of AAAC QSP and returned to NTDC(A)
- CPO2 Tallman reviewed changes by ADAC
- Submitted to NTDC(A) LSC TDO (Lt Meyers and SLT Leblanc) for final review

04 April 2022

- NTDC(A) LSC TDO (Lt Meyers) completed review and completed the QSP Quality Control review
- Reviewed MESCR and DPGR Cover Letter
- Commenced submission of CDAR (Minor Amendment) process

ANNEX D – TASK ANALYSIS DIAGRAM

This diagram represents the recreation or original task analysis (scalar) used to define the job.



ANNEX E – COSTING INFORMATION

This table provides an estimated cost to deliver this training. Considered costs include Temporary Duty (TD), material, and personnel costs. TD costs include Rations & Quarters (R&Q) costs for relevant TE, and National Joint Counsel factors (meals & incidentals). Material costs are best estimates. Personnel costs are IAW the Cost Factors Manual for the current fiscal year for all instructors identified in Chapter 1 - Required Course Staffing.

TD Costs	**# Trainees + Instructors /**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
Airfare (Return)	8	$1,000.00	n/a	$8,000	$8000	$40,000.00
R&Q	8	$36.00	20	$5,760	$5,760	$28,800.00
Incidentals	8	$17.50	20	$2,800	$2,800	$14,000.00
Subtotal				**$16,560**	**$16,560**	**$82,800.00**
Material Costs	**Quantity**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PP&S (Estimate)	16	$50.00	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.000	20			
Computer requirements	16	$1,000.00	1	$16,000	$0	$16,000.00
IT Equip (Smart board, etc.)	1	$6,000.00	1	$6,000	$0	$6,000.00
DWAN connection (Setup)	1	$5,000.00	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.00	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.00	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
Personnel Costs	**Quantity**	**Cost/day**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PO2 or Civilian Instructor	1	$###.##	###	$8,920	$8,920	$44,600.00
Subtotal				**$8,920**	**$8,920**	**$44,600.00**
Total Estimated Costs				**$53,745**	**$26,745**	**$133,725**

ANNEX F – ASSESSMENT PLAN

This table summarizes assessment (EC & PC) details contained within this QSP.

EC/PC	Scope	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess trainee's competence in Conducting Passive Aural Listening.	Practical	Trainee must obtain: 75% on both Analysis and Classification
EC 001.02	EO 001.02	To assess trainee's competence in Conducting Acoustic Gram Analysis.	Practical	Trainee must obtain: 75% on both Analysis and Classification
EC 001.03	EO 001.03	To assess trainee's competence in Conducting Tactical Analysis.	Practical	Trainee must obtain: 75% on both Analysis and Classification
PC 001	PO 001	To assess trainee's competence in Analyzing Acoustic Data.	Practical	Trainee must obtain: 75% on both Analysis and Classification

ANNEX G – TEL DEVELOPMENT OPPORTUNITIES

If a training methodology is determined to be ineffective and/or inefficient, a Learning Support Request (LSR) may be submitted to correct the deficiency. Technology Enabled Learning (TEL) development opportunities are assessed IAW NPTORD 4500-84 following submission.

Title	Requirement Description	Lesson #	LSR #

ANNEX H – SCHEDULING INFORMATION

This table summarizes the conditions cell in each LG. The order within the table reflects the order of instruction. All timings are based on maximum load. This is required for the Scheduling Cell and completion is mandatory.

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
PO 001 – ANALYZE ACOUSTIC DATA											
LG 001.02.01	150	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers
LG 001.01.01	450	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers
LG 001.01.02	100	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.01.03	250	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.01.04	250	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
EC 001.01	150	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers
LG 001.02.01	50	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers
LG 001.03.01	100	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers
LG 001.02.02	600	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.02.03	300	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.03.02	50	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.02.04	300	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.03.03	50	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers

LG 001.02.05	350	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.03.04	50	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.02.06	300	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.03.05	50	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
LG 001.02.07	2600	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers
LG 001.03.06	150	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY Laptop, Speakers
EC 001.03	100	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers
EC 001.02	300	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers
PC 001	300	ADAC or Classroom	Secret	1	13	1	1	13	1	Yes	OSPREY laptop, speakers

RECOMMENDED SCHEDULE

TIME	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
Period 1 0800-0850	**Course Introduction & Administration (BAT)**	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (PAL Lecture)	**LG 001.01.02** – Recognizing Aural Transient Events (PAL Lecture)	**LG 001.01.04** – Classifying contacts Aurally (PAL Labs)	**LG 001.03.01** – Interpret Tactical Events (Lecture/Demo)
Period 2 0850-0940	**Course Introduction & Administration (TAT)**	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (PAL Lecture)	**LG 001.01.03** – Analyzing Aural Acoustic Data (PAL Lecture/Lab)	**LG 001.01.04** – Classifying contacts Aurally (PAL Labs)	**LG 001.03.01** – Interpret Tactical Events (Lecture/Demo)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**LG 001.02.01** – Conduct Analysis Approach (Lecture)	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (PAL Lecture)	**LG 001.01.03** – Analyzing Aural Acoustic Data (PAL Lecture/Lab)	**LG 001.01.04** – Classifying contacts Aurally (PAL Labs)	**LG 001.02.02** – Identify Acoustic Sources (Lecture)
Period 4 1040-1130	**LG 001.02.01** – Conduct Analysis Approach (Lecture)	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (PAL Lecture)	**LG 001.01.03** – Analyzing Aural Acoustic Data (PAL Lecture/Lab)	**LG 001.01.04** – Classifying contacts Aurally (PAL Labs)	**LG 001.02.02** – Identify Acoustic Sources (Lecture)
1130-1230	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
Period 5 1230-1320	**LG 001.02.01** – Conduct Analysis Approach (ARS Lecture)	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (PAL Lecture)	**LG 001.01.03** – Analyzing Aural Acoustic Data (PAL Lecture/Lab)	**EC 001.01** – Conduct PAL	**LG 001.02.02** – Identify Acoustic Sources (Lecture)
Period 6 1320-1410	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (OSPREY Lecture)	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (PAL Lecture)	**LG 001.01.03** – Analyzing Aural Acoustic Data (PAL Lecture/Lab)	**EC 001.01** – Conduct PAL	**LG 001.02.02** – Identify Acoustic Sources (Lecture)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (PAL Lecture)	**LG 001.01.02** – Recognizing Aural Transient Events (PAL Lecture)	**LG 001.01.04** – Classifying contacts Aurally (PAL Labs)	**LG 001.02.01** – Conduct Analysis Approach (Guest Lecture)	**LG 001.02.02** – Identify Acoustic Sources (Lecture)
Period 8 1510-1600	**LG 001.01.01** – Recognizing Aural Acoustic Characteristics (PAL Lecture)	**PT**	**PT**	**EC 001.01** – Conduct PAL (Debrief)	**PT**

TIME	DAY 6	DAY 7	DAY 8	DAY 9	DAY 10
Period 1 0800-0850	**LG 001.02.02** – Identify Acoustic Sources (Lecture)	**LG 001.02.03** – Analyzing Diesel Submarines (Ice Breaker/Lecture)	**LG 001.02.04** – Analyzing Non-Allied Nuclear Submarines (Ice Breaker/Lecture)	**NO FOREIGN** **LG 001.02.05** – Analyzing Allied Nuclear Submarines (Ice Breaker/Lecture)	**LG 001.02.06** – Analyzing Naval Surface Vessels (Ice Breaker/Lecture)
Period 2 0850-0940	**LG 001.02.02** – Identify Acoustic Sources (Lecture)	**LG 001.02.03** – Analyzing Diesel Submarines (Lecture)	**LG 001.02.04** – Analyzing Non-Allied Nuclear Submarines (Lecture)	**NO FOREIGN** **LG 001.02.05** – Analyzing Allied Nuclear Submarines (Lecture)	**LG 001.02.06** – Analyzing Naval Surface Vessels (Lecture)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**LG 001.02.02** – Identify Acoustic Sources (Lecture)	**LG 001.02.03** – Analyzing Diesel Submarines (Lecture/Practical Lab)	**LG 001.02.04** – Analyzing Non-Allied Nuclear Submarines (Lecture/Practical Lab)	**NO FOREIGN** **LG 001.02.05** – Analyzing Allied Nuclear Submarines (Lecture/Practical Lab)	**LG 001.02.06** – Analyzing Naval Surface Vessels (Practical Lab)
Period 4 1040-1130	**LG 001.02.02** – Identify Acoustic Sources (Lecture)	**LG 001.02.03** – Analyzing Diesel Submarines (Practical Lab)	**LG 001.02.04** – Analyzing Non-Allied Nuclear Submarines (Practical Lab)	**NO FOREIGN** **LG 001.02.05** – Analyzing Allied Nuclear Submarines (Practical Lab)	**LG 001.02.06** – Analyzing Naval Surface Vessels (Practical Lab)
1130-1230	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 5 1230-1320	**LG 001.02.02** – Identify Acoustic Sources (Lecture)	**LG 001.02.03** – Analyzing Diesel Submarines (Practical Lab)	**LG 001.02.04** – Analyzing Non-Allied Nuclear Submarines (Practical Lab)	**NO FOREIGN** **LG 001.02.05** – Analyzing Allied Nuclear Submarines (Practical Lab)	**LG 001.02.06** – Analyzing Naval Surface Vessels (Practical Lab)
Period 6 1320-1410	**LG 001.02.02** – Identify Acoustic Sources (Lab)	**LG 001.03.02** – Tactical Analysis of Diesel Submarines (Quick Kill)	**LG 001.03.03** – Tactical Analysis of Non-Allied Nuclear Submarines (Quick Kill)	**NO FOREIGN** **LG 001.02.05** – Analyzing Allied Nuclear Submarines (Practical Lab)	**LG 001.03.05** – Tactical Analysis of Naval Surface Vessels (Quick Kill)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**LG 001.02.02** – Identify Acoustic Sources (Lab)	**LG 001.02.03** – Analyzing Diesel Submarines (Lab Debrief)	**LG 001.02.04** – Analyzing Non-Allied Nuclear Submarines (Lab Debrief)	**NO FOREIGN** **LG 001.03.04** – Tactical Analysis of Allied Nuclear Submarines (Quick Kill)	**LG 001.02.06** – Analyzing Naval Surface Vessels (Lab Debrief)
Period 8 1510-1600	**PT**	**PT**	**PT**	**NO FOREIGN** **LG 001.02.05** – Analyzing Allied Nuclear Submarines (Lab Debrief)	**PT**

TIME	DAY 11	DAY 12	DAY 13	DAY 14	DAY 15
Period 1 0800-0850	**LG 001.02.07** – Mixed Lab #1 (Ice Breaker/Lab)	**LG 001.02.07** – Mixed Lab #2 (Ice Breaker/Lab)	**LG 001.02.07** – Mixed Lab #3 (Ice Breaker/Lab)	**LG 001.02.07** – Mixed Lab #4 (Ice Breaker/Lab)	**LG 001.02.07** – Mixed Lab #5 (Ice Breaker/Lab)
Period 2 0850-0940	**LG 001.02.07** – Mixed Lab #1 (Practical Lab)	**LG 001.02.07** – Mixed Lab #2 (Practical Lab)	**LG 001.02.07** – Mixed Lab #3 (Practical Lab)	**LG 001.02.07** – Mixed Lab #4 (Practical Lab)	**LG 001.02.07** – Mixed Lab #5 (Practical Lab)
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**LG 001.02.07** – Mixed Lab #1 (Practical Lab)	**LG 001.02.07** – Mixed Lab #2 (Practical Lab)	**LG 001.02.07** – Mixed Lab #3 (Practical Lab)	**LG 001.02.07** – Mixed Lab #4 (Practical Lab)	**LG 001.02.07** – Mixed Lab #5 (Practical Lab)
Period 4 1040-1130	**LG 001.02.07** – Mixed Lab #1 (Practical Lab)	**LG 001.02.07** – Mixed Lab #2 (Practical Lab)	**LG 001.02.07** – Mixed Lab #3 (Practical Lab)	**LG 001.02.07** – Mixed Lab #4 (Practical Lab)	**LG 001.02.07** – Mixed Lab #5 (Practical Lab)
1130-1230	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 5 1230-1320	**LG 001.02.07** – Mixed Lab #1 (Practical Lab)	**LG 001.02.07** – Mixed Lab #2 (Practical Lab)	**LG 001.02.07** – Mixed Lab #3 (Practical Lab)	**LG 001.02.07** – Mixed Lab #4 (Practical Lab)	**LG 001.02.07** – Mixed Lab #6 (Practical Lab)
Period 6 1320-1410	**LG 001.02.07** – Mixed Lab #1 (Practical Lab)	**LG 001.02.07** – Mixed Lab #2 (Practical Lab)	**LG 001.02.07** – Mixed Lab #3 (Practical Lab)	**LG 001.02.07** – Mixed Lab #4 (Practical Lab)	**LG 001.02.07** – Mixed Lab #6 (Practical Lab)
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**LG 001.03.06** – Rapid Tactical Analysis Mixed Lab (Quick Kill)	**LG 001.03.06** – Rapid Tactical Analysis Mixed Lab (Quick Kill)	**LG 001.03.06** – Rapid Tactical Analysis Mixed Lab (Quick Kill)	**EC 001.03 – Conduct Tactical Analysis** (Quick Kill)	**LG 001.02.07** – Mixed Lab #6 (Practical Lab)
Period 8 1510-1600	**LG 001.02.07** – Day 1 Mixed Lab (Lab Debrief)	**LG 001.02.07** – Day 2 Mixed Lab (Lab Debrief)	**LG 001.02.07** – Day 3 Mixed Lab (Lab Debrief)	**LG 001.02.07** – Day 4 Mixed Lab (Lab Debrief)	**LG 001.02.07** – Mixed Lab #6 (Practical Lab/Debrief)

TIME	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20
Period 1 0800-0850	**LG 001.02.07** – Mixed Lab #7 (Ice Breaker/Lab)	**EC 001.02 –** Conduct Acoustic Gram Analysis	**LG 001.02.07** – Mixed Lab #9 (Ice Breaker/Lab)	**PC 001**	**PT**
Period 2 0850-0940	**LG 001.02.07** – Mixed Lab #7 (Practical Lab)	**EC 001.02 –** Conduct Acoustic Gram Analysis	**LG 001.02.07** – Mixed Lab #9 (Practical Lab)	**PC 001**	**PT**
0940-0950	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 3 0950-1040	**LG 001.02.07** – Mixed Lab #7 (Practical Lab)	**EC 001.02 –** Conduct Acoustic Gram Analysis	**LG 001.02.07** – Mixed Lab #9 (Practical Lab)	**PC 001**	**TAT (Course Critique /Reports)**
Period 4 1040-1130	**LG 001.02.07** – Mixed Lab #7 (Practical Lab)	**EC 001.02 –** Conduct Acoustic Gram Analysis	**LG 001.02.07** – Mixed Lab #9 (Practical Lab)	**PC 001**	**BAT (Graduation)**
1130-1230	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 5 1230-1320	**LG 001.02.07** – Mixed Lab #8 (Practical Lab)	**EC 001.03 – Conduct Tactical Analysis** (Debrief)	**LG 001.02.07** – Mixed Lab #10 (Practical Lab)	**PT**	**Out Routine**
Period 6 1320-1410	**LG 001.02.07** – Mixed Lab #8 (Practical Lab)	**PT**	**LG 001.02.07** – Mixed Lab #10 (Practical Lab)	**PT**	**Out Routine**
1410-1420	**BREAK**	**BREAK**	**BREAK**	**BREAK**	**BREAK**
Period 7 1420-1510	**LG 001.02.07** – Mixed Lab #8 (Practical Lab)	**EC 001.02 –** Conduct Acoustic Gram Analysis (Debrief)	**LG 001.02.07** – Mixed Lab #10 (Practical Lab)	**PC 001 (Debrief)**	**Out Routine**
Period 8 1510-1600	**LG 001.02.07** – Mixed Lab #8 (Practical Lab/Debrief)	**EC 001.02 –** Conduct Acoustic Gram Analysis (Debrief)	**LG 001.02.07** – Mixed Lab #10 (Practical Lab/Debrief)	**PC 001 (Debrief)**	**Out Routine**

ANNEX J – EVALUATION MATRIX

General
The following guidance is provided to assist in the development and maintenance of the Advanced Acoustic Analysis student evaluation.

Concept of Summative Testing
Summative testing is used to determine whether the student has achieved the PO and/or Critical EO and is used at the end of a phase of instruction.

The following PC and Critical EC will be assessed cumulatively:

Test Name	**PC 001**
Objective	Conduct Acoustic Gram Analysis
Evaluation Method	Practical
Description, Written	N/A
Description, Practical	The test will contain 6 mixed contacts with varying degrees of difficulty.
Time	4 x 50 min period
Pass/Fail Criteria	Pass Mark: 75% for analysis and 75% for classification.
Contribution to Final Pass/Fail Assessment	Yes.
Contribution to Course Final Mark	Represents 50% of the final grade Weighting: 80% of analysis mark, 20% of classification

When	Upon completion of EO 001.01, 001.02 and 001,03
Resources	Workstation Acoustic data files
Design Responsibility	TRINITY (ADAC)
Approving Authority	TRINITY (ADAC)

Test Name	**EC 001.02**
Objective	Conduct Acoustic Gram Analysis
Evaluation Method	Practical
Description, Written	N/A
Description, Practical	The test will contain 6 mixed contacts with varying degrees of difficulty.
Time	4 x 50 min period
Pass/Fail Criteria	Pass Mark: 75% for analysis and 75% for classification.
Contribution to Final Pass/Fail Assessment	Yes.
Contribution to Course Final Mark	Represents 20% of the final grade Weighting: 80% of analysis mark, 20% of classification mark.

When	Upon completion of EO 001.01, 001.02 and 001,03
Resources	Workstation Acoustic data files
Design Responsibility	TRINITY (ADAC)
Test Name	EC 001.02
Approving Authority	TRINITY (ADAC)

Concept of Formative Testing
Formative testing is used to help the student and the instructor to recognize progress or lapses in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods.

The following EOs will be tested formatively:

Test Name	**EC 001.01**
Objective	Conduct PAL
Evaluation Method	Practical
Description, Written	N/A

Description, Practical	This assessment shall consist of a minimum of 20 sources representing a cross section of the material instructed. Delivery by instructor led or computer based module. Students are required to answer questions based on the sound source given.
Time	2 x 50 minute period
Pass/Fail Criteria	75%
Contribution to Final Pass/Fail Assessment	No
Contribution to Course Final Mark	Represents 10% of the final grade
When	Upon completion of EO 001.01
Resources	Workstation Audio files
Design Responsibility	TRINITY (ADAC)
Approving Authority	TRINITY (ADAC)

Test Name	**EC 001.03**
Objective	Conduct Tactical Analysis
Evaluation Method	Practical
Description, Written	N/A

Description, Practical	This assessment consists of 10 contacts that represent a cross section of the material instructed. Delivery by instructor led or computer based module. Students are to answer questions provided that pertain to the example given.
Time	1 x 50 minute period.
Pass/Fail Criteria	75%
Contribution to Final Pass/Fail Assessment	No .
Contribution to Course Final Mark	Represents 20% of the final grade
When	Upon completion of EO 001.01, EO 001.02 and EO 001.03.
Test Name	EC 001.03
Resources	Workstation Acoustic data files
Design Responsibility	TRINITY (ADAC)
Approving Authority	TRINITY (ADAC)